

ATG6 interacting with NPR1 increases *Arabidopsis thaliana* resistance to *Pst* DC3000/*avrRps4* by increasing its nuclear accumulation and stability

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Abstract

Autophagy-related gene 6 (ATG6) plays a crucial role in plant immunity. Nonexpressor of pathogenesis-related genes1 (NPR1) acts as a signaling hub of plant immunity. However, the relationship between ATG6 and NPR1 is unclear. Here, we find that ATG6 directly interacts with NPR1. *ATG6* overexpression significantly increased nuclear accumulation of NPR1. Furthermore, we demonstrate that *ATG6* increases NPR1 protein levels and improves its stability. Interestingly, ATG6 promotes the formation of SINC (SA-induced NPR1 condensates)-like condensates. Additionally, ATG6 and NPR1 synergistically promote the expression of *pathogenesis-related* genes. Further results showed that silencing *ATG6* in *NPR1-GFP* exacerbates *Pst* DC3000/*avrRps4* invasion, while double overexpression of *ATG6* and *NPR1* synergistically inhibits *Pst* DC3000/*avrRps4* invasion. In summary, our findings unveil an interplay of NPR1 with ATG6 and elucidate important molecular mechanisms for enhancing plant immunity.

Highlight

We unveil a novel relationship in which ATG6 positively regulates NPR1 in plant immunity.

eLife assessment

This **valuable** study details an aspect of plant immunity where ATG6 was not previously known to have a role. The results suggest a direct relationship between ATG6 and NPR1, a well-studied salicylic acid receptor protein, which could be of interest to researchers studying the regulation of plant immunity. While the data presented are **compelling**, there are concerns about the interpretation of results, particularly regarding discrepancies in fluorescence and protein blot data. Addressing these issues would improve the overall impact of the work and consistency with prior studies.

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Introduction

Plants are constantly challenged by pathogens in nature. In order to survive and reproduce, plants have evolved complex mechanisms to cope with attack by pathogens (Jones and Dangl, 2006). Nonexpressor of pathogenesis-related genes 1 (NPR1) is a key regulator of plant immunity (Chen *et al.*, 2021b). It contains the BTB/POZ (Broad Complex, Tramtrack, and Bric-a-brac/Pox virus and Zinc finger) domain in the N-terminal region, the ANK (Ankyrin repeats) domain in the middle region, and SA-binding domain (SBD) and the nuclear localization sequence (NLS) in the C-terminal region (Cao *et al.*, 1997; Rochon *et al.*, 2006; Kumar *et al.*, 2022). NPR1 is a receptor of SA (salicylic acid) mainly localized as an oligomer in the cytoplasm and sensitive to the surrounding redox state (Tada *et al.*, 2008; Wu *et al.*, 2012). SA mediates the dynamic oligomer to dimer response of NPR1 (Tada *et al.*, 2008) and promotes translocation of NPR1 into the nucleus, which increases plant resistance to pathogens by activating the expression of immune-related genes (Kinkema *et al.*, 2000; Chen *et al.*, 2021b).

NPR1 is mainly degraded by the ubiquitin proteasome system (UPS) (Spoel *et al.*, 2009; Saleh *et al.*, 2015; Skelly *et al.*, 2019). An increasing researches have shown that autophagy and the UPS pathway play overlapping roles in regulating intracellular protein homeostasis (Zhou *et al.*, 2014; Marshall *et al.*, 2015; Kikuchi *et al.*, 2020). Our previous study showed that ATGs (autophagy-related genes) are involved in NPR1 turnover (Gong *et al.*, 2020). Autophagy negatively regulates *Pst* DC3000/*avrRpm1*-induced programmed cell death (PCD) via the SA receptor NPR1 (Yoshimoto *et al.*, 2009). These results imply that ATGs might be involved in plant immunity through the regulation of NPR1 homeostasis. However, the detailed mechanism has not yet been elucidated.

ATG6 is the homologues of yeast Vps30/Atg6 and mammalian BECN1/Beclin1 (Xu *et al.*, 2017). It is a common and required subunit of the class III phosphatidylinositol 3-kinase (PtdIns3K) lipid kinase complexes, which regulates autophagosome nucleation in *Arabidopsis thaliana* (*Arabidopsis*) (Qi *et al.*, 2017; Wang *et al.*, 2020). The homozygous *atg6* mutant is lethal, suggesting that ATG6 is essential for plant growth and development (Fujiki *et al.*, 2007; Qin *et al.*, 2007; Harrison-Lowe and Olsen, 2008; Patel and Dinesh-Kumar, 2008). In *Arabidopsis*, *N. benthamiana* and wheat, ATG6 or its homologues was reported to act as a positive regulator to enhance plant disease resistance to *P. syringae* pv. *tomato* (*Pst*) DC3000 and *Pst* DC3000/*avrRpm1* bacteria (Patel and Dinesh-Kumar, 2008), *N. benthamiana* mosaic virus (TMV) (Liu *et al.*, 2005), turnip mosaic virus (TuMV) (Li *et al.*, 2018), pepper mild mottle virus (PMMoV) (Jiao *et al.*, 2020), and *Blumeria graminis* f. sp. *tritici* (*Bgt*) fungus (Yue *et al.*, 2015). Several research teams have also elucidated that ATG6 interacted with Bax Inhibitor-1 (NbBI-1) (Xu *et al.*, 2017).

and RNA-dependent RNA polymerase (RdRp) (Li *et al.*, 2018) to suppress pathogen invasion. However, the mechanism by which ATG6 suppresses pathogen invasion by regulating NPR1 has not yet been reported.

Here, we show that ATG6 and NPR1 synergistically enhance *Arabidopsis* resistance to *Pst* DC3000/*avrRps4* infiltration. We discover that ATG6 increases NPR1 protein levels and nuclear accumulation of NPR1. Moreover, ATG6 can stabilize NPR1 and promote the formation of SINC (SA-induced NPR1 condensates)-like condensates. Our study revealed a unique mechanism in which NPR1 cooperatively increases plant immunity with ATG6.

Results

NPR1 physically interacts with ATG6 in vitro and in vivo

To examine the relationship between ATGs and NPRs, we predicted that some ATGs might interact with NPRs. In a yeast two-hybrid (Y2H) screen, we identified that NPR1, NPR3 and NPR4 could interact with ATG6 and several ATG8 isoforms (Fig. S1 and Results S1 in the Supplemental Data 1). In this study, we mainly investigated the relationship between ATG6 and NPR1 during the process of plant immune response. Firstly, the NPR1 truncations NPR1-N (1-328AA, containing the BTB/POZ domain, ANK1, ANK2,) and NPR1-C (328-594AA, containing the ANK3, ANK4, SA-binding domain (SBD) and NLS) were used to identify the interaction domains between NPR1 and ATG6. The results showed that NPR1-C interacted with full-length ATG6 in yeast (Fig. 1a, line 3). The interaction between NPR1 and SnRK2.8 was used as a positive control (Lee *et al.*, 2015). Secondly, pull-down assays were performed *in vitro*. NPR1-His bound GST-ATG6, but not GST (Fig. 1b). Furthermore, co-immunoprecipitation (Co-IP) assays were performed in *N. benthamiana*, as shown in Fig. 1c, ATG6-mCherry was co-immunoprecipitated with NPR1-GFP. In Fig. S2, fluorescence signals of NPR1-GFP and ATG6-mCherry were co-localized in the nucleus under normal and 1 mM SA treatment conditions. The interaction between ATG6 and NPR1 was also verified by a bimolecular fluorescence complementation (BiFC) assay (Fig. 1d). These results demonstrate that ATG6 interacts with NPR1 both *in vitro* and *in vivo*.

ATG6 co-localized with NPR1 in the nucleus

Remarkably, under normal and SA treatment conditions, we found that ATG6 is localized in the cytoplasm and nucleus, and it co-localized with NPR1 in the nucleus (Fig. S2). Nuclear localization of ATG6 was also observed in *N. benthamiana* transiently transformed with ATG6-mCherry and ATG6-GFP under normal and SA treatment conditions (Fig. 2a and b). ATG6-GFP co-localizes with the nuclear localization marker nls-mCherry (indicated by white arrows) (Fig. 2b). Additionally, we observed punctate patterns indicative of canonical autophagy-like localization of ATG6-GFP fluorescence signals (indicated by red circles) (Fig. 2b). The nuclear localization signal of ATG6 was also observed in *UBQ10::ATG6-GFP* overexpressing *Arabidopsis* (Fig. S3a). To exclude the possibility that the observed localization of ATG6-GFP is due to free GFP, ATG6-GFP and free GFP levels in *UBQ10::ATG6-GFP Arabidopsis* and *N. benthamiana* were detected before and after SA treatment.

Notably, no free GFP was detected and this means that the fluorescence signal observed by confocal microscopy is ATG6-GFP, not free GFP (Fig. S3d and e).

Moreover, the nuclear and cytoplasmic fractions were separated. Under SA treatment, ATG6-mCherry and ATG6-GFP were detected in the cytoplasmic and nuclear fractions in *N. benthamiana* (Fig. 2c and d). Similarly, ATG6 was also detected in the nuclear fraction of *UBQ10::ATG6-GFP* and *UBQ10::ATG6-mCherry* overexpressing plants, and SA promoted both cytoplasm and nuclear accumulation of ATG6 (Fig. 2e and f). Additionally, we obtained ATG6 and NPR1 double

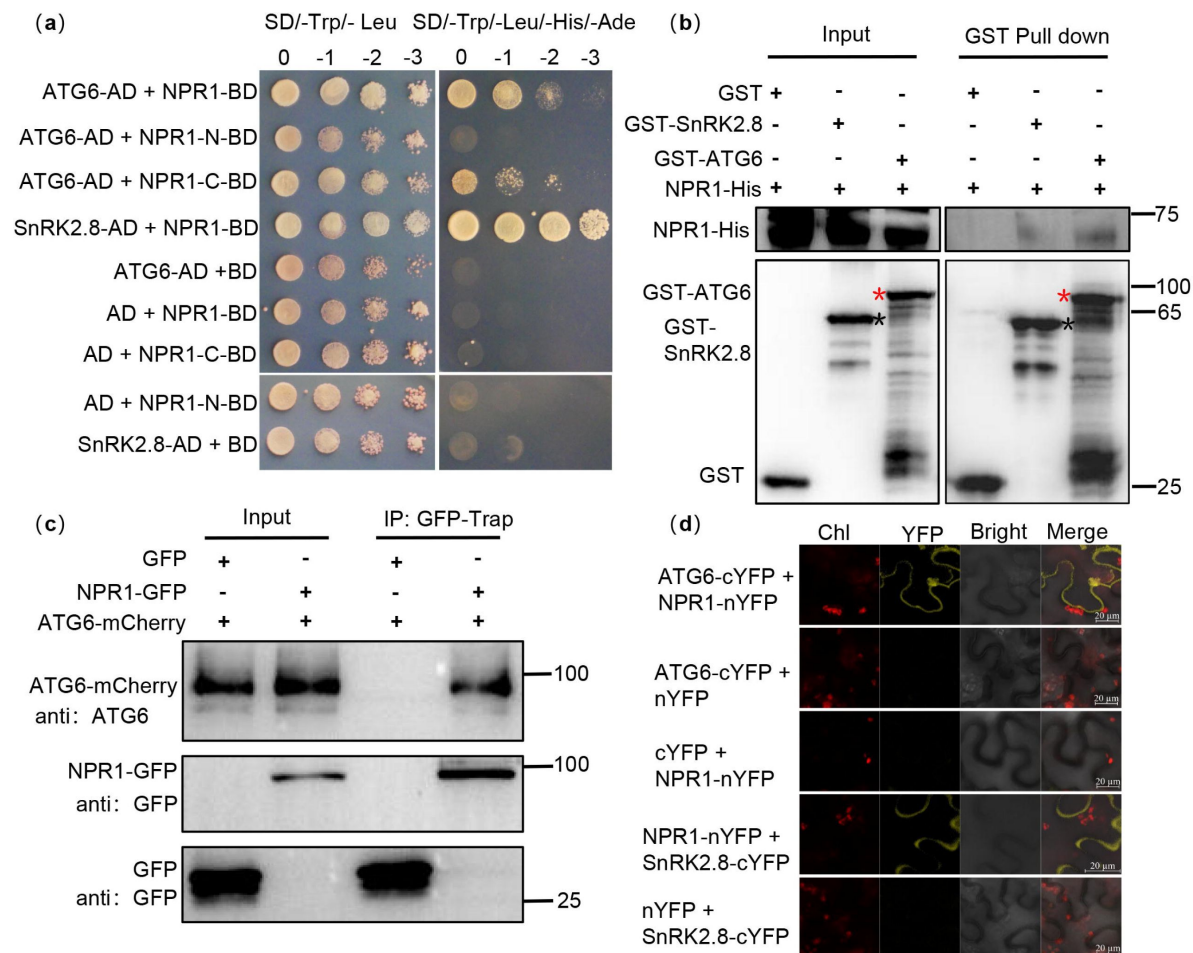


Figure 1.

Physical interaction between NPR1 and ATG6.

(a). Interaction of NPR1 with ATG6 in yeast. The CDS of *ATG6*, *NPR1*, *NPR1-N* (1~984 bp), *NPR1-C* (984~1782 bp) and *SnRK2.8* were fused to pGADT7 (AD) and pGBKT7 (BD), respectively. Co-transformation of NPR1-BD + AD, BD + ATG6-AD, BD + SnRK2.8-AD, NPR1-N-BD + AD, NPR1-C-BD + AD were used as negative controls. The interaction of NPR1-BD and SnRK2.8-AD was used as a positive control. Yeast growth on SD/-Trp-Leu-His-Ade media represents interaction. Numbers represent the dilution fold of yeast. 0, -1 (10 fold dilution), -2 (100 fold dilution), -3 (1000 fold dilution). (b). In vitro pull-down assays of NPR1-His with GST-ATG6 fusion protein. NPR1-His prokaryotic proteins were incubated with GST-tag Purification Resin conjugated with GST-ATG6, GST and SnRK2.8-GST. Western blotting analysis with anti-GST and anti-His. Black asterisk indicate SnRK2.8-GST bands. Red asterisk indicate GST-ATG6 bands. (c). Co-immunoprecipitation of NPR1 with ATG6 in vivo. Total protein was extracted from *N. benthamiana* transiently transformed with ATG6-mCherry + GFP and ATG6-mCherry + NPR1-GFP, followed by IP with GFP-Trap. Western blots analysis with ATG6 and GFP antibodies. (d). Bimolecular fluorescence complementation assay of NPR1 with ATG6 in *Nicotiana benthamiana* leaves. Agrobacterium carrying ATG6-nYFP and NPR1-cYFP was co-expressed in leaves of *Nicotiana benthamiana* for 3 days. As a positive control, NPR1-nYFP and SnRK2.8-cYFP were co-expressed. As negative controls, nYFP and ATG6-cYFP, NPR1-nYFP and cYFP, nYFP and SnRK2.8-cYFP were co-expressed. Confocal images were obtained from chloroplast (Chl), YFP, Bright field. Scale bar = 20 μm. All experiments were performed with three biological replicates.

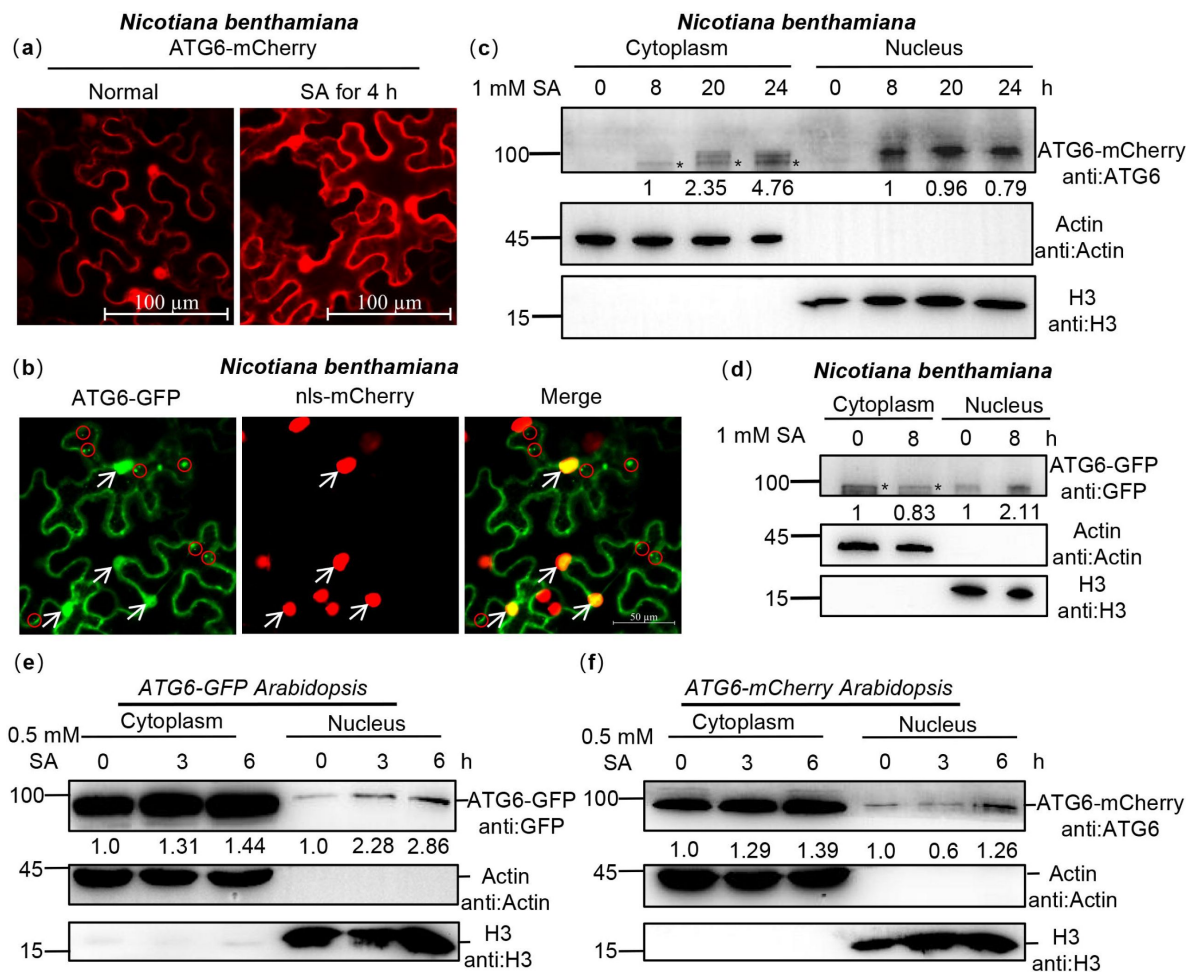


Figure 2.

ATG6 is localized in the cytoplasm and nucleus.

(a). The nuclear localization of ATG6-mCherry in *N. benthamiana*. Scale bar, 100 μ m. (b). Co-localization of ATG6-GFP and nls-mCherry in *N. benthamiana*. Scale bar, 50 μ m. (c). Subcellular fractionation of ATG6-mCherry in *N. benthamiana* after 1 mM SA treatment. Black asterisk (*) indicate ATG6-mCherry bands. (d). Subcellular fractionation of ATG6-GFP in *N. benthamiana* after 1 mM SA treatment. Black asterisk (*) indicate ATG6-GFP bands. (e). Subcellular fractionation of ATG6-GFP in *ATG6-GFP Arabidopsis* after 0.5 mM SA treatment. (f). Subcellular fractionation of ATG6-mCherry in *ATG6-mCherry Arabidopsis* after 0.5 mM SA treatment. In (c-f), ATG6-mCherry (c and f) and ATG6-GFP (d and e) were detected using ATG6 or GFP antibody. Actin and H3 were used as cytoplasmic and nucleus internal reference, respectively. Numerical values indicate quantitative analysis of ATG6-mCherry and ATG6-GFP using image J. All experiments were performed with three biological replicates.

overexpression of *Arabidopsis* *UBQ10::ATG6-mCherry* × *35S::NPR1-GFP* (*ATG6-mCherry* × *NPR1-GFP*) by crossing and screening (Fig. S4a). In *ATG6-mCherry* × *NPR1-GFP*, we observed co-localization of *ATG6-mCherry* with *NPR1-GFP* in the nucleus (Fig. S3b). These results are consistent with the prediction of the subcellular location of *ATG6* in the *Arabidopsis* subcellular database (<https://suba.live/>) (Fig. S3c). Additionally, we have conducted an investigation into the localization of endogenous *ATG6* in Col. Our results demonstrate that endogenous *ATG6* is present in both the nucleus and cytoplasm, and we have observed that SA treatment promotes the accumulation of *ATG6* in the nucleus (Fig. S5). Together, these findings suggest that *ATG6* is localized to both cytoplasm and nucleus, and co-localized with *NPR1* in the nucleus.

ATG6 overexpression increased nuclear accumulation of NPR1

Previous studies have shown that the nuclear localization of *NPR1* is essential for improving plant immunity (Kinkema *et al.*, 2000; Chen *et al.*, 2021b). We observed that a stronger nuclear localization signal of *NPR1-GFP* in *ATG6-mCherry* × *NPR1-GFP* leaves than that in *NPR1-GFP* under normal condition and 0.5 mM SA treatment for 3 h (Fig. 3a-b and Fig. S6). These findings indicate that *ATG6* might increase nuclear accumulation of *NPR1*. To exclude the possibility that the observed localization of *NPR1-GFP* is due to free GFP, we detected the levels of *NPR1-GFP* and free GFP in *ATG6-mCherry* × *NPR1-GFP* plants before and after SA treatment. Only ~ 10 % of free GFP was detected in *ATG6-mCherry* × *NPR1-GFP* plants before and after SA treatment, confirming that the observed localization of *NPR1-GFP* is not due to free GFP (Fig. S4b). Furthermore, the nuclear and cytoplasmic fractions of *ATG6-mCherry* × *NPR1-GFP* and *NPR1-GFP* were separated. Under normal conditions, the nuclear fractions *NPR1-GFP* in *ATG6-mCherry* × *NPR1-GFP* and *NPR1-GFP* were relatively weaker (Fig. 3c), which differs from the above observation (Fig. 3a). We speculate that this phenomenon might be attributed to the rapid turnover of *NPR1* in the nucleus (Spoel *et al.*, 2009; Saleh *et al.*, 2015). Consistent with the fluorescence distribution results, the nuclear fractions of *NPR1-GFP* in *ATG6-mCherry* × *NPR1-GFP* were significantly higher than those in *NPR1-GFP* under 0.5 mM SA treatment for 3 and 6 h (Fig. 3c). Furthermore, *Agrobacterium* harboring *ATG6-mCherry* and *NPR1-GFP* were transiently transformed to *N. benthamiana* leaves. After 1 d, the leaves were treated with 1 mM SA for 8 and 20 h. Subsequently nucleoplasmic separation experiments were performed. Similar to *Arabidopsis*, increased nuclear accumulation of *NPR1* was found when *ATG6* was overexpressed (Fig. 3e). Notably, we found that the ratio (nucleus *NPR1*/total *NPR1*) in *ATG6-mCherry* × *NPR1-GFP* was not significantly different from that in *NPR1-GFP* after SA treatment, and a similar phenomenon was observed in *N. benthamiana* (Fig. 3d and f). These results suggested that the increased nuclear accumulation of *NPR1* in *ATG6-mCherry* × *NPR1-GFP* plants might attributed to higher levels and more stable *NPR1* rather than the enhanced nuclear translocation of *NPR1* facilitated by *ATG6*. Furthermore, we validated the functionality of the *ATG6-GFP* and *ATG6-mCherry* fusion proteins utilized in this study by examining the phenotypes of *ATG6-GFP* and *ATG6-mCherry Arabidopsis* plants under carbon starvation conditions (Fig. S7 and Results S2 in the Supplemental Data 1).

ATG6 increases endogenous SA levels and promotes the expression of NPR1 downstream target genes

NPR1 localized in the nucleus is essential for activation of immune gene expression (Kinkema *et al.*, 2000; Chen *et al.*, 2021b). In our study, we observed that *ATG6* overexpression increased nuclear accumulation of *NPR1* (Fig. 3) and demonstrated an interaction between *ATG6* and *NPR1* in the nucleus (Fig. 1d). Therefore, we speculate that *ATG6* might regulate *NPR1* transcriptional activity. Notably, the expression level of *ICS1* in *ATG6-mCherry* × *NPR1-GFP* seedlings was significantly higher than that in *NPR1-GFP* under normal and SA treatment conditions (Fig. S8). Free SA levels in *ATG6-mCherry* × *NPR1-GFP* were also significantly higher compared to *NPR1-GFP* under *Pst* DC3000/*avrRps4* treatment. While there was no significant difference was observed under normal condition (Fig. 4a), this may be related to free SA consumption, as it can be converted to bound SA (Ding and Ding, 2020). In addition, the

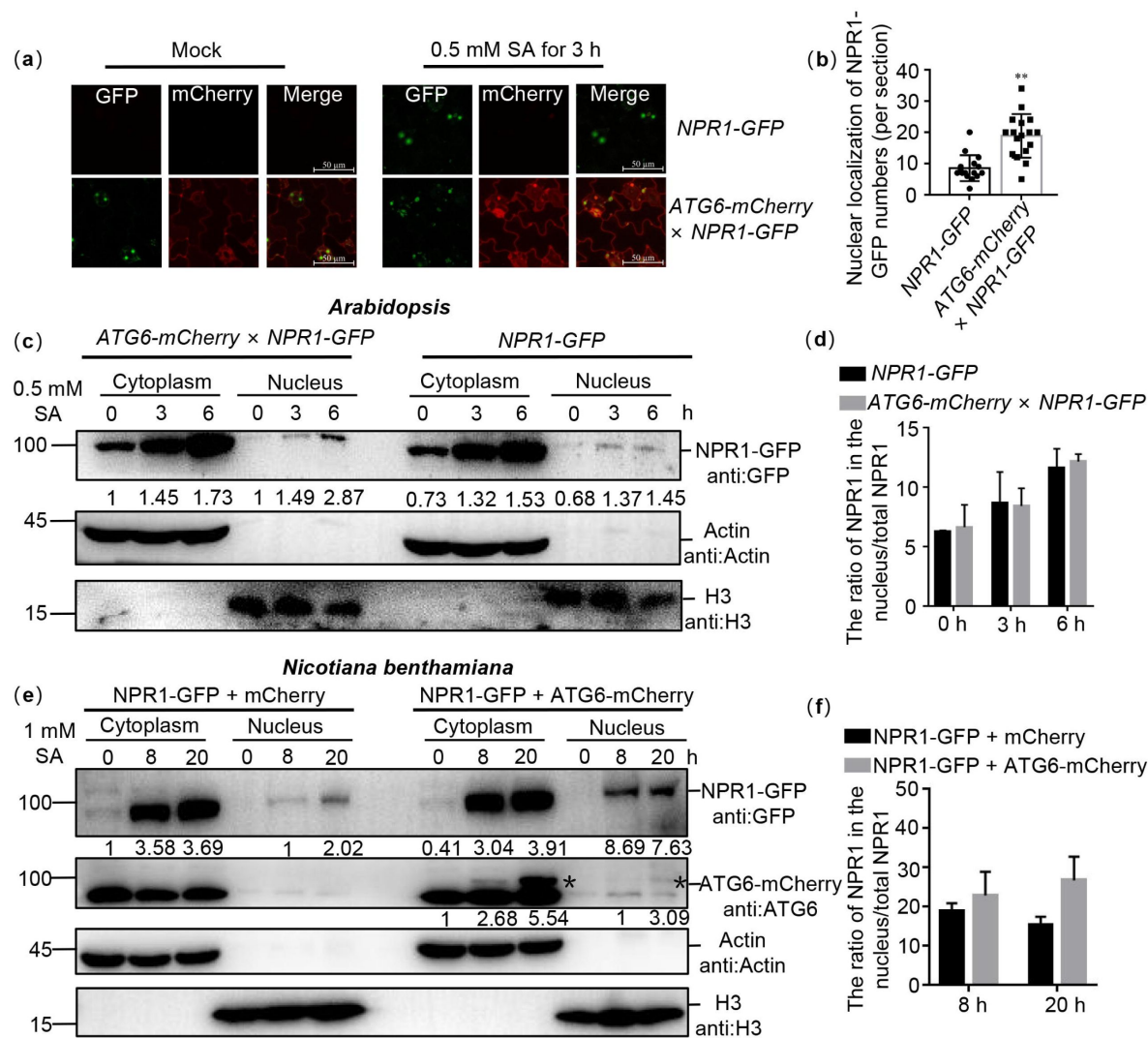


Figure 3.

ATG6 increases the nuclear accumulation of NPR1 under SA treatment.

(a). Confocal images of NPR1-GFP nuclear localization in 7-day-old seedlings of *NPR1-GFP* and *ATG6-mCherry × NPR1-GFP* under normal and 0.5 mM SA spray for 3 h. Scale bar, 50 μ m. (b). The count of nuclear localizations of NPR1-GFP in *ATG6-mCherry × NPR1-GFP* and *NPR1-GFP* Arabidopsis plants following SA treatment. In (a), Statistical data were obtained from three independent experiments, each comprising five individual images, resulting in a total of 15 images analyzed for this comparison. (c). Subcellular fractionation of NPR1-GFP in 7-day-old seedlings of *NPR1-GFP* and *ATG6-mCherry × NPR1-GFP* after 0.5 mM SA treatment for 0, 3 and 6 h. (d). The ratio of NPR1 in the nucleus/total NPR1 in (c). (e). Subcellular fractionation of NPR1-GFP in *N. benthamiana* after 1 mM SA treatment for 0, 8 and 20 h. (f). The ratio of NPR1 in the nucleus/total NPR1 in (e). In (c and e), cytoplasmic and nuclear proteins were extracted from *Arabidopsis* or *N. benthamiana*. NPR1-GFP were detected using GFP antibody. Actin and H3 were used as cytoplasmic and nucleus internal reference, respectively. Numerical values indicate quantitative analysis of NPR1-GFP using image J. All experiments were performed with three biological replicates.

expression of *PR1* (pathogenesis-related gene) and *PR5* in *ATG6-mCherry* × *NPR1-GFP* was significantly higher than that of *NPR1-GFP* under normal and SA treatment conditions (Fig. 4b and c). The expression of *PR1* and *PR5* in *ATG6-mCherry* was significantly higher than that of Col under *Pst* DC3000/*avrRps4* treatment (Fig. S9). These results support the role of ATG6 in facilitating the expression of NPR1 downstream *PR1* and *PR5* genes.

ATG6 increases NPR1 protein levels and the formation of SINC-like condensates

Interestingly, similar to previous reports (Zavaliev *et al.*, 2020), SA promoted the translocation of NPR1 into the nucleus, but still a significant amount of NPR1 was present in the cytoplasm (Fig. 3c and e). Previous studies have shown that SA increased NPR1 protein levels and facilitated the formation of SINC-like condensates in the cytoplasm, which are known to promote cell survival (Zavaliev *et al.*, 2020). In our experiments, we observed that under SA treatment, the protein levels of NPR1 in *ATG6-mCherry* × *NPR1-GFP* were significantly higher than that in *NPR1-GFP* (Fig. 5a). To further support our conclusions, we proceeded to silence *ATG6* in *NPR1-GFP* (*NPR1-GFP/silencing ATG6*) and subsequently assessed the protein level of NPR1-GFP before and after SA treatment. Our findings revealed that the protein level of NPR1-GFP in *NPR1-GFP/silencing ATG6* under SA treatment was notably lower than that in the *NPR1-GFP/Negative control* (Fig. S10). Under SA treatment, the protein levels of NPR1-GFP in *N. benthamiana* co-transformed with *ATG6-mCherry* + *NPR1-GFP* were also significantly higher than that of *mCherry* + *NPR1-GFP* (Fig. 5b). We also detected the expression of *NPR1* was detected. It is worth noting that NPR1 up-regulation was more obvious in Col after 3 h treatment with *Pst* DC3000/*avrRps4*. After 6 h treatment with *Pst* DC3000/*avrRps4*, there was no significant difference in the expression of NPR1 between Col and *ATG6-mCherry* (Fig. S11). These results suggest that ATG6 increases NPR1 protein levels. After SA treatment, more SINC-like condensates fluorescence were observed in *N. benthamiana* co-transformed with *ATG6-mCherry* + *NPR1-GFP* compared to *mCherry* + *NPR1-GFP* (Fig. 5c-d and Supplemental movie 1-2). Additionally, we observed that SINC-like condensates signaling partial co-localized with certain ATG6-mCherry autophagosomes fluorescence signals (Fig. S12). Taken together, these results suggest that ATG6 increases the protein levels of NPR1 and promotes the formation of SINC-like condensates, possibly caused by ATG6 increasing SA levels *in vivo*.

ATG6 maintains the protein stability of NPR1

Maintaining the stability of NPR1 is critical for enhancing plant immunity (Skelly *et al.*, 2019). To further verify whether ATG6 regulates NPR1 stability, we co-transfected *NPR1-GFP* with *ATG6-mCherry* or *mCherry* in *N. benthamiana* and performed cell-free degradation assays. Our results showed that *NPR1-GFP* degradation was significantly delayed when *ATG6* was overexpressed (Fig. S13). A similar trend was observed in *Arabidopsis*, where the *NPR1-GFP* protein in *ATG6-mCherry* × *NPR1-GFP* showed a slower degradation rate compared to *NPR1-GFP* during 0~180 min time period in a cell-free degradation assay (Fig. 6a and b). Moreover, when *Arabidopsis* seedlings were treated with cycloheximide (CHX) to block protein synthesis, we found that *NPR1-GFP* in *NPR1-GFP* was degraded after CHX treatment for 3~9 h and the half-life of *NPR1-GFP* is ~3 h, while the half-life of *NPR1-GFP* in *ATG6-mCherry* × *NPR1-GFP* is ~9 h (Fig. 6c and d). In addition, we also analyzed the degradation of *NPR1-GFP* in *NPR1-GFP* and *NPR1-GFP/atg5* following 100 μM cycloheximide (CHX) treatment. The results show that the degradation rate of *NPR1-GFP* in *NPR1-GFP/atg5* plants was similarly to that in *NPR1-GFP* plants (Fig. 6e and f). These results indicate that ATG6 plays a role in maintaining the stability of NPR1, which may also be related to the fact that ATG6 promotes an increase in free SA *in vivo*, since SA has the function of increasing NPR1 stability (Ding *et al.*, 2016; Skelly *et al.*, 2019).

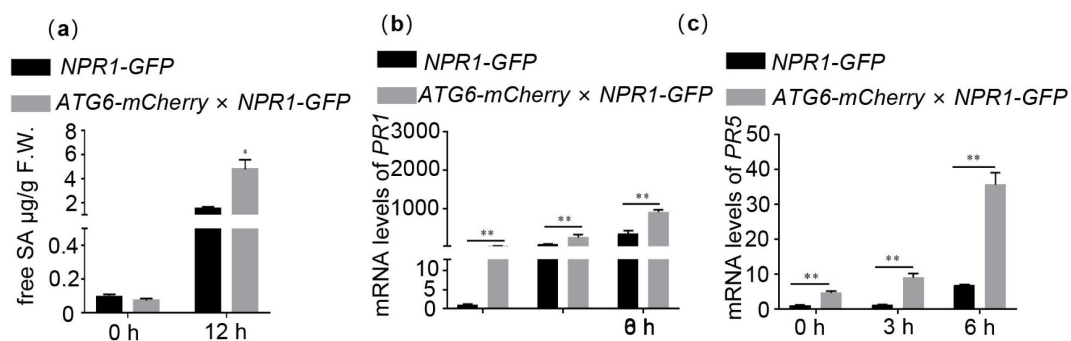


Figure 4.

ATG6 increases endogenous SA levels and promotes the expression of NPR1 downstream target genes

(a). Level of free SA in 3-week-old *NPR1-GFP* and *ATG6-mCherry × NPR1-GFP* after *Pst* DC3000/*avrRps4* for 12 h. (b-c). Expression of *PR1* in 3-week-old *NPR1-GFP* and *ATG6-mCherry × NPR1-GFP* (b), *PR5* (c) under normal and SA treatment conditions. * or ** indicates that the significant difference compared to the control is at the level of 0.05 or 0.01 (Student *t* test *p* value, * *p* < 0.05 or ** *p* < 0.01). All experiments were performed with three biological replicates.

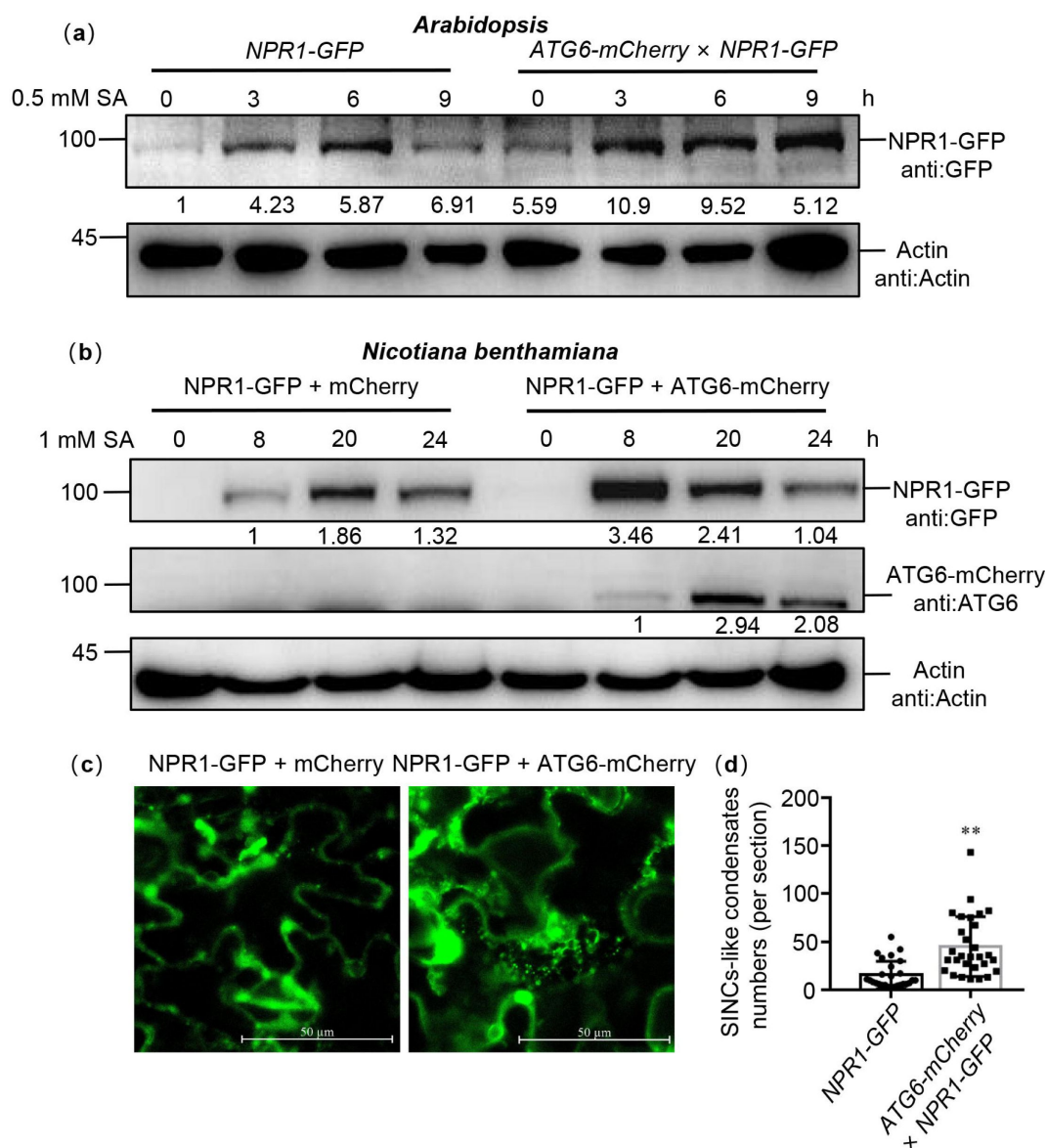


Figure 5.

ATG6 increases the NPR1 protein levels and the formation of SINC-like condensates.

(a). NPR1-GFP protein levels in 7-day-old seedlings of *NPR1-GFP* and *ATG6-mCherry × NPR1-GFP* after 0.5 mM SA treatment for 0, 3, 6 and 9 h. Numerical values indicate quantitative analysis of NPR1-GFP protein using image J. (b). NPR1-GFP protein levels in *N. benthamiana*. ATG6-mCherry + NPR1-GFP, NPR1-GFP + mCherry were co-expressed in *N. benthamiana*. After 2 days, leaves were treated with 1 mM SA for 8, 20, 24 h. Total proteins were extracted and analyzed. Numerical values indicate quantitative analysis of NPR1-GFP protein using image J. (c). ATG6 promotes the formation of SINC-like condensates. ATG6-mCherry + NPR1-GFP, NPR1-GFP + mCherry were co-expressed in *N. benthamiana*. After 2 days, leaves were treated with 1 mM SA for 24 h. Confocal images obtained at excitation with wavelengths of 488 nm, scale bar = 50 μ m. (d). SINC-like condensates numbers of per section in (c), $n > 10$ sections. ** indicates that the significant difference compared to the control is at the level of 0.01 (Student *t* test *p* value, ** $p < 0.01$). All experiments were performed with three biological replicates.

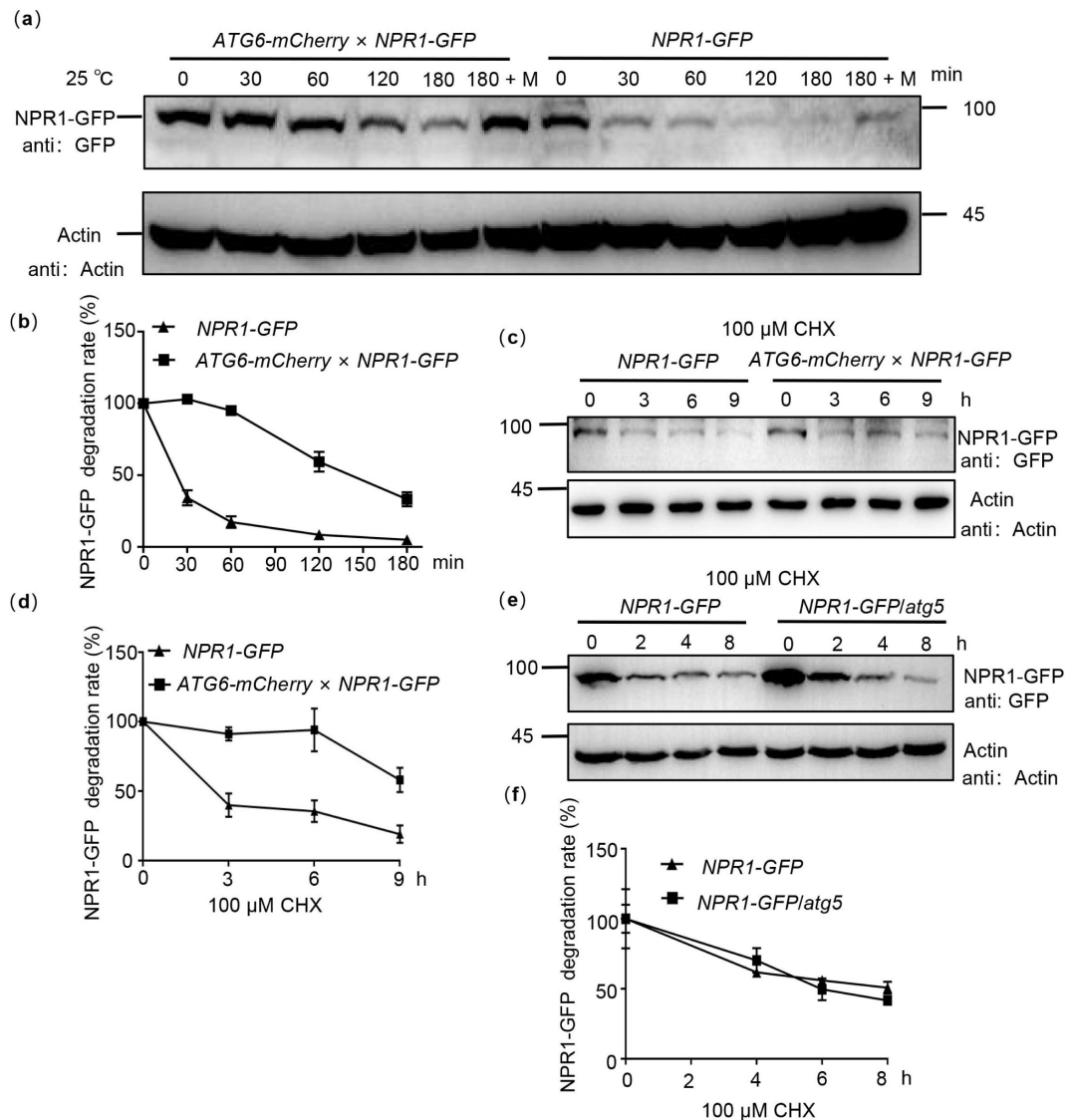


Figure 6.

ATG6 improves the protein stability of NPR1.

(a). NPR1-GFP degradation assay in *Arabidopsis*. Total proteins from 7-day-old seedlings of *NPR1-GFP* and *ATG6-mCherry × NPR1-GFP* were extracted, using Actin as an internal reference. "M" indicates MG115 treatment. **(b).** Quantification of NPR1-GFP degradation rates in **(a)** using Image J. In **(a)** and **(b)**, the extracts were incubated for 0 ~ 180 min at room temperature (25°C), the degradation rate of NPR1-GFP was analyzed. **(c).** NPR1-GFP protein turnover. 7-day-old *NPR1-GFP* and *ATG6-mCherry × NPR1-GFP* seedlings were treated with 100 μM cycloheximide (CHX) for different times. Total proteins were analyzed, Actin was used as an internal reference. **(d).** Quantification of NPR1-GFP protein turnover rates in **(c)** using Image J. **(e).** NPR1-GFP protein turnover. 7-day-old *NPR1-GFP* and *NPR1-GFP/atg5* seedlings were treated with 100 μM cycloheximide (CHX) for different times. Total proteins were analyzed, Actin was used as an internal reference. **(f).** Quantification of protein levels of NPR1-GFP in **(e)** using Image J. All experiments were performed with three biological replicates.

ATG6 and NPR1 cooperatively inhibit invasion of *Pst* DC3000/*avrRps4*

The mRNA expression levels of *ATG6* in Col were significantly increased after 6 h, 12 h and 24 h under *Pst* DC3000/*avrRps4* treatment (Fig. 7a). Similarly, both the *ATG6* gene and protein were significantly up-regulated under 0.5 mM SA treatment (Fig. 7b and c). These results suggest that the expression of *ATG6* could be induced by *Pst* DC3000/*avrRps4* and 0.5 mM SA treatment.

Considering that *ATG6* increases NPR1 protein levels (Fig. 5a-b) and promotes its nuclear accumulation (Fig. 3), as well as maintains NPR1 stability (Fig. 6), then we studied the role of *ATG6*-NPR1 interactions in plant immunity. However, studying the function of *ATG6* is challenging due to the lethality of homozygous *atg6* mutant (Qin *et al.*, 2007; Harrison-Lowe and Olsen, 2008; Patel and Dinesh-Kumar, 2008). According to our previous report (Lei *et al.*, 2020; Zhang *et al.*, 2023), *ATG6* was silenced using artificial miRNA^{ATG6} (amiRNA^{ATG6}) delivered by the gold nanoparticles (AuNPs). First, the effect of *ATG6* silencing in Col on the plant immune response was investigated. Similar to *atg5*, Col/silencing *ATG6* exhibited more active growth of *Pst* DC3000/*avrRps4* than Col/negative control (NC) after *Pst* DC3000/*avrRps4* infiltration for 3 days (Fig. 7d). Furthermore, according to the previously reported methods (Ohira *et al.*, 2017; Gomez *et al.*, 2022), we generated two amiRNA^{ATG6} lines (amiRNA^{ATG6} # 1 and amiRNA^{ATG6} # 2) designed against *ATG6* and placed under the control of a β -estradiol inducible promoter. There were no significant phenotypic differences in amiRNA^{ATG6} # 1 compared to the Col, while amiRNA^{ATG6} # 2 exhibited a slight leaf developmental defect (Fig. 7e and f). Subsequently, we investigated the expression of *ATG6* following treatment with 100 μ M β -estradiol. Our results showed that, after 100 μ M β -estradiol treatment for 1~3 d, the expression of *ATG6* in both amiRNA^{ATG6} # 1 and amiRNA^{ATG6} # 2 lines was significantly lower than that in Col. Specifically, the expression of *ATG6* in the amiRNA^{ATG6} #1 and amiRNA^{ATG6} #2 lines decreased by 50~70% compared with Col (Fig. 7g). Furthermore, to assess the function of *ATG6* in plant immune, we performed infiltrations of *Pst* DC3000/*avrRps4* after 100 μ M β -estradiol treatment for 24 h. We compared the growth of *Pst* DC3000/*avrRps4* in the amiRNA^{ATG6} lines and Col. The results clearly demonstrate that the growth of *Pst* DC3000/*avrRps4* in amiRNA^{ATG6} # 1 and amiRNA^{ATG6} # 2 was significantly more compared to Col (Fig. 7h). Moreover, we silenced *ATG6* in *NPR1-GFP* (*NPR1-GFP*/silencing *ATG6*), and *NPR1-GFP/atg5* (crossed *NPR1-GFP* with *atg5* to obtain *NPR1-GFP/atg5*) was used as an autophagy-deficient control. There was more *Pst* DC3000/*avrRps4* growth in *NPR1-GFP*/silencing *ATG6* and *NPR1-GFP/atg5* compared to *NPR1-GFP*/NC after *Pst* DC3000/*avrRps4* infiltration (Fig. 7i). In contrast, the growth of *Pst* DC3000/*avrRps4* in *NPR1-GFP*, *ATG6-mCherry*, *ATG6-mCherry* $\times$ *NPR1-GFP* was significantly lower than that in Col and *npr1* (Fig. 7j) and was the lowest in *ATG6-mCherry* $\times$ *NPR1-GFP* (Fig. 7j).

These results confirm that *ATG6* and *NPR1* cooperatively enhance *Arabidopsis* resistance to inhibit *Pst* DC3000/*avrRps4* invasion. Together, these results suggest that *ATG6* improves plant resistance to pathogens by regulating *NPR1*.

Discussion

Although SA signaling and autophagy are related to the plant immune system (Yoshimoto *et al.*, 2009; Munch *et al.*, 2014; Wang *et al.*, 2016), the connection of these two processes in plant immune processes and their interaction is rarely reported. Previous studies have shown that unrestricted pathogen-induced PCD requires SA signalling in autophagy-deficient mutants. SA and its analogue benzo (1,2,3) thiadiazole-7-carbothioic acid (BTH) induce autophagosome production (Yoshimoto *et al.*, 2009). Moreover, autophagy has been shown to negatively regulates *Pst* DC3000/*avrRpm1*-induced PCD via the SA receptor *NPR1* (Yoshimoto *et al.*, 2009), implying that

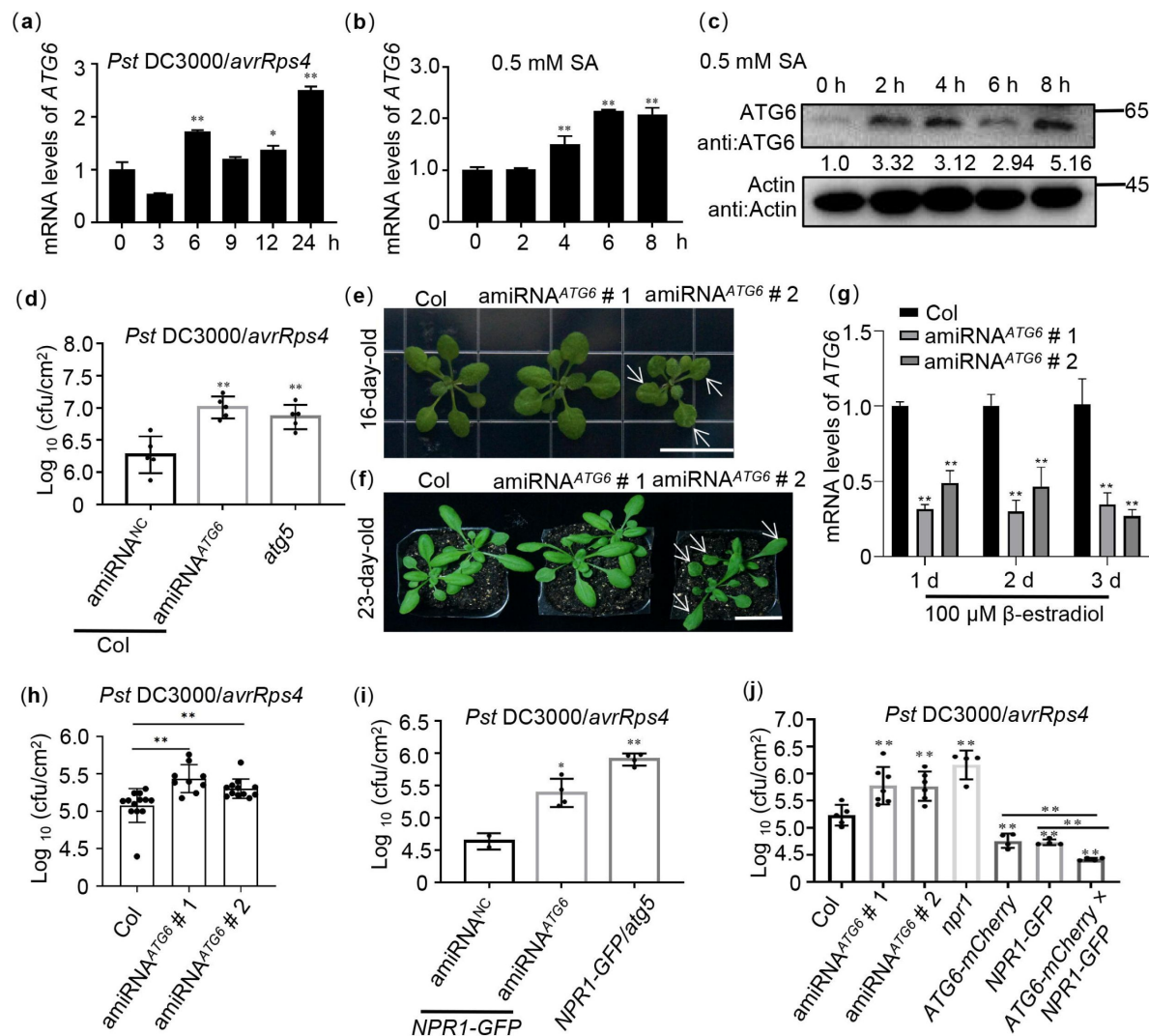


Figure 7.

ATG6 and NPR1 cooperatively inhibit the growth of *Pst* DC3000/*avrRps4*.

(a). Expression of ATG6 under *Pst* DC3000/*avrRps4* infiltration in 3-week-old Col leaves. (b). Expression of ATG6 in the presence of 0.5 mM SA in 3-week-old Col leaves. (c). The protein levels of ATG6 after 0.5 mM SA in 3-week-old Col leaves. Total leaf proteins from *Arabidopsis* were analyzed, Actin was used as an internal reference. Numerical values indicate quantitative analysis of ATG6 protein using image J. (d). Growth of *Pst* DC3000/*avrRps4* in Col/silencing ATG6 and Col/negative control (NC). (e). Phenotypes of 16-day-old *amiRNA*^{ATG6} #1 and *amiRNA*^{ATG6} #2. Bar, 1 cm. (f). Phenotypes of 23-day-old *amiRNA*^{ATG6} #1 and *amiRNA*^{ATG6} #2. Bar, 3 cm. (g). Expression of ATG6 in Col, *amiRNA*^{ATG6} #1 and *amiRNA*^{ATG6} #2 under infiltration treatment of 100 μM β-estradiol. (h). Growth of *Pst* DC3000/*avrRps4* in *Arabidopsis* leaves of *amiRNA*^{ATG6} #1, *amiRNA*^{ATG6} #2 and Col. (i). Growth of *Pst* DC3000/*avrRps4* in *NPR1*-GFP/silencing ATG6 and *NPR1*-GFP/NC. (j). Growth of *Pst* DC3000/*avrRps4* in *Arabidopsis* leaves of Col, *amiRNA*^{ATG6} #1, *amiRNA*^{ATG6} #2, *npr1*, *NPR1*-GFP, *ATG6*-mCherry and *ATG6*-mCherry × *NPR1*-GFP. In (d, h-j), a low dose of *Pst* DC3000/*avrRps4* (OD₆₀₀ = 0.001) was infiltrated. After 3 days, the growth of *Pst* DC3000/*avrRps4* was counted. * or ** indicates that the significant difference compared to the control is at the level of 0.05 or 0.01 (Student *t* test *p* value, * *p* < 0.05 or ** *p* < 0.01). All experiments were performed with three biological replicates.

autophagy regulates SA signaling through a negative feedback loop to limit immune-related PCD. Here, we demonstrated that ATG6 increases NPR1 protein levels and nuclear accumulation (**Fig. 3** and **5**). Additionally, ATG6 also maintains the stability of NPR1 and promotes the formation of SINC-like condensates (**Fig. 5** and **6**). These findings introduce a novel perspective on the positive regulation of NPR1 by ATG6, highlighting their synergistic role in enhancing plant resistance.

Our results confirmed that *ATG6* overexpression significantly increased nuclear accumulation of NPR1 (**Fig. 3**). ATG6 also increases NPR1 protein levels and improves NPR1 stability (**Fig. 5** and **6**). Therefore, we consider that the increased nuclear accumulation of NPR1 in *ATG6-mCherry* × *NPR1-GFP* plants might result from higher levels and more stable NPR1 rather than the enhanced nuclear translocation of NPR1 facilitated by ATG6. To verify this possibility, we determined the ratio of NPR1-GFP in the nuclear localization versus total NPR1-GFP. Notably, the ratio (nucleus NPR1/total NPR1) in *ATG6-mCherry* × *NPR1-GFP* was not significantly different from that in *NPR1-GFP*, and there is a similar phenomenon in *N. benthamiana* (**Fig. 3c-f**). Further we analyzed whether ATG6 affects NPR1 protein levels and protein stability. Our results show that ATG6 increases NPR1 protein levels under SA treatment and ATG6 maintains the protein stability of NPR1 (**Fig. 5** and **6**). These results suggested that the increased nuclear accumulation of NPR1 by ATG6 result from higher levels and more stable NPR1.

NPR1 is an important signaling hub of the plant immune response. Nuclear localization of NPR1 is essential to enhance plant resistance (Kinkema *et al.*, 2000; Chen *et al.*, 2021b), it interacts with transcription factors such as TGAs in the nucleus to activate expression of downstream target genes (Chen *et al.*, 2019; Chen *et al.*, 2021a). A recent study showed that nuclear-located ATG8h recognizes C1, a geminivirus nuclear protein, and promotes C1 degradation through autophagy to limit viral infiltration in solanaceous plants (Li *et al.*, 2020). Here, we confirmed that ATG6 is also distributed in the nucleus and ATG6 is co-localized with NPR1 (**Fig. 1d** and **2**), suggesting that ATG6 interact with NPR1 in the nucleus. ATG6 synergistically inhibits the invasion of *Pst* DC3000/*avrRps4* with NPR1. Chen *et al.* found that in the nucleus, NPR1 can recruit enhanced disease susceptibility 1 (EDS1), a transcriptional coactivator, to synergistically activate expression of downstream target genes (Chen *et al.*, 2021a). Previous studies have shown that acidic activation domains (AADs) in transcriptional activators (such as Gal4, Gcn4 and VP16) play important roles in activating downstream target genes. Acidic amino acids and hydrophobic residues are the key structural elements of AAD (Pennica *et al.*, 1984; Cress and Triezenberg, 1991; Van Hoy *et al.*, 1993). Chen *et al.* found that EDS1 contains two ADD domains and confirmed that EDS1 is a transcriptional activator with AAD (Chen *et al.*, 2021a). Here, we also have similar results that *ATG6* overexpression significantly enhanced the expression of *PR1* and *PR5* (**Fig. 4b-c** and **Fig. S9**), and that the ADD domain containing acidic and hydrophobic amino acids is also found in ATG6 (148-295 AA) (**Fig. S14**). We speculate that ATG6 might act as a transcriptional coactivator to activate *PRs* expression synergistically with NPR1.

A recent study showed that SA not only enhances plant resistance by increasing NPR1 nuclear import and transcriptional activity, but also promotes cell survival by coordinating the distribution of NPR1 in the nucleus and cytoplasm (Zavaliev *et al.*, 2020). Notably, NPR1 accumulated in the cytoplasm recruits other immunomodulators (such as EDS1, PAD4 etc) to form SINC-like condensates to promote cell survival (Zavaliev *et al.*, 2020). Similarly, we also found that NPR1 accumulated abundantly in the cytoplasm after SA treatment and that ATG6 significantly increased NPR1 protein levels (**Fig. 3c**, **3e** and **5a-b**). Obviously, the accumulation of NPR1 in the cytoplasm may be related to ATG6 synergizing with NPR1 to enhance plant resistance. Interestingly, *ATG6* overexpression significantly increased the formation of SINC-like condensates (**Fig. 5c-d** and **Supplemental movie 1-2**), which should also be a way for ATG6 and NPR1 to synergistically resist invasion of pathogens. We consider that ATG6 promotes the formation of SINC-like condensates through the dual action of endogenous and exogenous SA. Considering that ATG6 promotes SINC-like condensates formation, we further examined changes in cell death in

Col, amiRNA^{ATG6} # 1, amiRNA^{ATG6} # 2, *npr1*, *NPR1-GFP*, *ATG6-mCherry* and *ATG6-mCherry* × *NPR1-GFP* plants. The results of Taipan blue staining showed that *Pst* DC3000/*avrRps4*-induced cell death in *npr1*, amiRNA^{ATG6} # 1 and amiRNA^{ATG6} # 2 was significantly higher compared to Col (**Fig. S15**). Conversely, *Pst* DC3000/*avrRps4*-induced cell death in *ATG6-mCherry*, *NPR1-GFP* and *ATG6-mCherry* × *NPR1-GFP* was significantly lower compared to Col. Notably, *Pst* DC3000/*avrRps4*-induced cell death in *ATG6-mCherry* × *NPR1-GFP* was significantly lower compared *ATG6-mCherry* and *NPR1-GFP* (**Fig. S15**). These results suggest that ATG6 and NPR1 cooperatively inhibit *Pst* DC3000/*avrRps4*-induced cell death.

ATG6 is a common and required subunit of PtdIns3K lipid kinase complexes, which regulates autophagosome nucleation in *Arabidopsis* (Qi *et al.*, 2017 [DOI](#); Bozhkov, 2018 [DOI](#)). In this study, we also found that ATG6 can maintain the stability of NPR1. Thus, to confirm whether the regulation of NPR1 protein stability by ATG6 is autophagy-dependent, we used autophagy inhibitors (Concanamycin A, ConA and Wortmannin, WM) to detect the degradation of NPR1-GFP. Cell-free degradation assays showed that 100 μM MG115 treatment significantly inhibited the degradation of NPR1-GFP. However, 5 μM concanamycin A treatment did not significantly delay NPR1 degradation (**Fig. S16**). Remarkably, treatment with 30 μM Wortmannin resulted in a slight acceleration of NPR1 degradation, while the combined treatment of ConA and WM significantly expedited the degradation of NPR1 (**Fig. S16**). This may be related to crosstalk between autophagy and 26S Proteasome. It has been demonstrated that autophagy directly regulates the activity of the 26S proteasome under normal conditions or treatment with *Pst* DC3000 (Marshall *et al.*, 2015 [DOI](#); Ustun *et al.*, 2018 [DOI](#)). Marshall *et al.* found that the 26S proteasome subunits (RPN1, RPN3, RPN5, RPN10, PAG1, PBF1) are significantly enriched in autophagy-deficient mutants under normal growth conditions (Marshall *et al.*, 2015 [DOI](#)). Treatment with concanamycin A (ConA), an inhibitor of vacuolar-type ATPase, increased the level of the 20S proteasome subunit PBA1 under treatment with *Pst* DC3000 (Ustun *et al.*, 2018 [DOI](#)). In addition, we also analyzed the degradation of NPR1-GFP in *NPR1-GFP* and *NPR1-GFP/atg5* following 100 μM cycloheximide (CHX) treatment. The results show that the degradation rate of NPR1-GFP in *NPR1-GFP/atg5* plants was similarly to that in *NPR1-GFP* plants (**Fig. 6e and f** [DOI](#)). These results suggest that deletion of ATG5 do not affect the protein stability of NPR1.

An increasing number of studies have shown that ATGs differentially affect plant immunity. Deletion of ATGs (ATG5, ATG7, ATG10 etc.) leads to reduced resistance of plants to necrotrophic pathogens (Lai *et al.*, 2011 [DOI](#); Lenz *et al.*, 2011 [DOI](#); Minina *et al.*, 2018 [DOI](#)). ATGs can directly interact with other proteins to positively regulate plant immunity. In *N. benthamiana*, ATG8f interacts the effector protein βC1 of the cotton *leaf curl multan virus* and promotes its degradation to limit pathogen invasion (Haxim *et al.*, 2017 [DOI](#)). Notably, ATG18a can interact with WRKY33 transcription factor to synergistically against *Botrytis* invasion (Lai *et al.*, 2011 [DOI](#)). Our evidence shows that ATG6 interacts with NPR1 and works together to counteract pathogen invasion by positively regulating NPR1 and SA levels *in vivo*. In conclusion, we unveil a novel relationship in which ATG6 positively regulates NPR1 in plant immunity (**Fig. 8** [DOI](#)). ATG6 interacts with NPR1 to synergistically enhance plant resistance by regulating NPR1 protein levels, stability, nuclear accumulation, and formation of SINC-like condensates.

Material and Methods

Plasmid construction

Details of plasmid construction methods are listed in Methods S1 of Supplemental Data 1, primer used are listed in Table S1 and S2 of Supplemental Data 1. The mapping of vectors is listed in Supplemental Data 2.

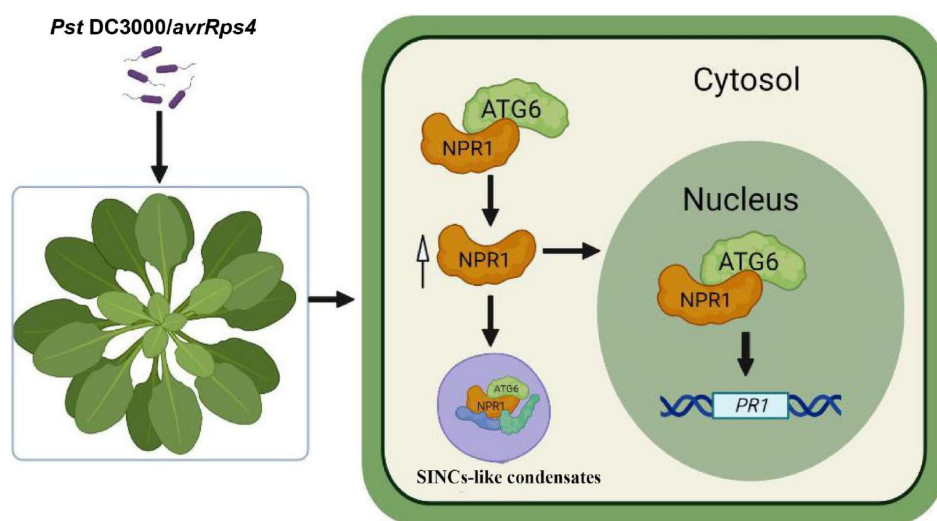


Figure 8.

Working model for NPR1 regulation by ATG6.

ATG6 interacts directly with NPR1 to increase NPR1 protein level and stability, thereby promoting the formation of SINC-like condensates and increasing the nuclear accumulation of NPR1. ATG6 synergistically activates *PRs* expression with NPR1 to cooperatively enhance resistance to inhibit *Pst* DC3000/avrRps4 invasion in *Arabidopsis*.

Plant material *Arabidopsis*

35S::NPR1-GFP (in *npr1-2* background) and *npr1-1* were kindly provided by Dr. Xinnian Dong of Duke University; *atg5-1* (SALK_020601).

UBQ10::ATG6-mCherry, UBQ10::ATG6-GFP and amiRNA^{ATG6} lines were obtained by *Agrobacterium* transformation (Clough and Bent, 1998 [\[1\]](#)). ATG6, NPR1 double overexpression of *Arabidopsis* (ATG6-mCherry × NPR1-GFP) and NPR1-GFP/*atg5* were obtained by crossing respectively.

Full description of the *Arabidopsis* screening is included Methods S2 of Supplemental Data 1. Details of plant material are listed in Table S3 of Supplemental Data 1.

Growth conditions *Arabidopsis thaliana*

All *Arabidopsis thaliana* (*Arabidopsis*) seeds were treated in 10% sodium hypochlorite for 7 min, washed with ddH₂O, and treated in 75% ethanol for 30 s, finally washed three times with ddH₂O. Seeds were sown in 1/2 MS with 2% sucrose solid medium, vernalized at 4°C for 2 days.

For 7-day-old *Arabidopsis* seedling cultures, the plates were placed under the following conditions: daily cycle of 16 h light (~80 μmol m⁻² s⁻¹) and 8 h dark at 23 ± 2°C.

For 3-week-old *Arabidopsis* cultures, after 7 days of growth on the plates, the seedlings were transferred to soil for further growth for 2 weeks under the same conditions (Zhang *et al.*, 2018 [\[2\]](#)).

N. benthamiana

For 3-week-old *N. benthamiana* cultures, seeds were sown in the soil and vernalized at 4°C for 2 days. After 10 days of growth on soil, the seedlings were transferred to soil for further growth for 2 weeks under the same conditions (Jiao *et al.*, 2019 [\[3\]](#)).

Treatment conditions

Treatment of 7-day-old seedlings

For SA treatment

7-day-old *Arabidopsis* seedlings were transferred to 1/2 MS liquid medium containing 0.5 mM SA for 0, 3 and 6 h, respectively. The corresponding results are shown in Fig. 2e [\[4\]](#), f, 3c [\[5\]](#), d and 5a.

For cycloheximide (CHX) treatment

Seedlings of *Arabidopsis* (7 days) were transferred to 1/2 MS liquid medium containing 100 μM CHX for 0, 3, 6 and 9 h, respectively. The corresponding results are shown in Fig. 6c and e [\[6\]](#).

Treatment of 3-week-old *Arabidopsis*

For silencing ATG6 in Col and NPR1-GFP

As previously described (Lei *et al.*, 2020 [\[7\]](#); Zhang *et al.*, 2022 [\[8\]](#); Zhang *et al.*, 2023 [\[9\]](#)), 1 mM gold nanoparticles (AuNPs) were synthesized. The artificial microRNA (amiRNA)^{ATG6} (UCAAUUCUAGGAUAACUGCCC) was designed based on the Web MicroRNA Designer (<http://wmd3.weigelworld.org/> [\[10\]](#)) platform. The complementary sequence of amiRNA^{ATG6} is located on the eighth exon of the ATG6 gene. The sequence of “UUCUCCGAACGUGUCACGUTT” was used as a

negative control (NC). NC is a universal negative control without species specificity (Gao *et al.*, 2018 [DOI](#); Lei *et al.*, 2020 [DOI](#)). amiRNA^{ATG6} and amiRNA^{NC} synthesized by Suzhou GenePharma. AuNPs (1 mM) and amiRNA^{ATG6} (20 μM) were incubated at a 9:1 ratio for 30 min at 25°C, 50 rpm. After incubation, a mixture of AuNPs and amiRNA^{ATG6} was diluted 15-fold with the infiltration buffer (pH 5.7, 10 mM MES, 10 mM MgCl₂) and infiltrated through the abaxial leaf surface into 3-week-old Col or *NPR1-GFP* for 1-3 days. The third day was chosen as material for *ATG6* silencing. After the third day of AuNPs-amiRNA^{ATG6} and AuNPs-amiRNA^{NC} infiltration, *Pst* DC3000/*avrRps4* was infiltrated, and then growth of *Pst* DC3000/*avrRps4* was detected.

For β-estradiol treatment

100 μM β-estradiol was infiltrated to treat 3-week-old *Arabidopsis* leaves. After 24 h of treatment with β-estradiol, *Pst* DC3000/*avrRps4* was infiltrated and then growth of *Pst* DC3000/*avrRps4* was detected after 3 d.

For *Pst* DC3000/*avrRps4* infiltration

Infiltration with *Pst* DC3000/*avrRps4* was performed as previously described (Wang *et al.*, 2016 [DOI](#); Skelly *et al.*, 2019 [DOI](#)). Full description of the *Pst* DC3000/*avrRps4* culture is included in Methods S3 of Supplemental Data 1.

For SA treatment

For 3-week-old Col, 0.5 mM SA was infiltrated into the leaves for 0, 2, 4, 6 and 8 h. The corresponding results are shown in **Fig. 7b and c** [DOI](#).

Yeast two-hybrid assay

Yeast two-hybrid experiments were performed according to the previously described protocol (Fu *et al.*, 2012 [DOI](#)). Full description of Yeast two-hybrid is included in Methods S4 of Supplemental Data 1.

Pull down assays in vitro

500 μL of GST, GST-ATG6, SnRK2.8-GST were incubated with GST-tag Purification Resin (Beyotime, P2250) for 2 h at 4°C. The mixture was then centrifuged at 1500 g for 1 min at 4°C, and the resin was washed three times with PBS buffer. Next, the GST-tag purification resin was incubated with the NPR1-His for 2 h at 4°C. After washing three times with PBS buffer, 2 × sample buffers were added to the resin and denatured at 100 °C for 10 mins. The resulting samples were then used for western blotting analysis. Full description of prokaryotic proteins expression is included in Methods S5 of Supplemental Data 1.

Co-immunoprecipitation

0.5 g leaves of *N. benthamiana* transiently transformed with ATG6-mCherry + GFP and ATG6-mCherry + NPR1-GFP were fully ground in liquid nitrogen and homogenized in 500 μL of lysis buffer (50 mM Tris-HCl pH 7.5, 150 mM NaCl, 0.5 mM EDTA, 5% Glycerol, 0.2% NP40, 1 mM PMSF, 40 μM MG115, protease inhibitor cocktail 500× and phosphatase inhibitor cocktail 5000×). The samples were then incubated on ice for 30 mins, and centrifuged at 10142 g (TGL16, cence, hunan, China) for 15 mins at 4°C. The supernatant (500 μL) was incubated with 20 μL of GFP-Trap Magnetic Agarose beads (ChromoTek, gtma-20) in a 1.5 mL Eppendorf tube for 2 h by rotating at 4°C. After incubation, the GFP-Trap magnetic Agarose beads were washed three times with cold wash buffer (50 mM Tris-HCl pH 7.5, 150 mM NaCl, 0.5 mM EDTA) and denatured at 75°C for 10 minutes after adding 2 × sample buffer. Western blotting was performed with antibodies to ATG6 and GFP.

Nuclear and cytoplasmic separation

Nuclear and cytoplasmic separation were performed according to the previously described method (Kinkema *et al.*, 2000 [DOI](#)). Full description of nuclear and cytoplasmic separation is given in Methods S6 of Supplemental Data 1.

Protein degradation in vitro

Protein degradation assays were performed according to a previously described method (Spoel *et al.*, 2009 [DOI](#); Saleh *et al.*, 2015 [DOI](#)). Full description of protein degradation is included in Methods S7 of Supplemental Data 1.

Protein Extraction and Western Blotting Analysis

Protein extraction and western blotting were performed as previously described (Lei *et al.*, 2020 [DOI](#); Zhang *et al.*, 2022 [DOI](#)). Protein was denatured at 100°C for 10 mins. NPR1 protein was denatured at 75°C for 10 mins (Lei *et al.*, 2020 [DOI](#)). Full description is included in Methods S8 of Supplemental Data 1. Antibody information is presented in Table S4 of Supplemental Data 1.

Confocal microscope observation

For nuclear localization of NPR1-GFP observation

7-day-old seedlings of *NPR1-GFP* and *ATG6-mCherry* × *NPR1-GFP* were sprayed with 0.5 mM SA for 0 and 3 h. GFP and mCherry fluorescence signals in leaves were observed under the confocal microscope (Zeiss LSM880). Statistical data were obtained from three independent experiments, each comprising five individual images, resulting in a total of 15 images analyzed for this comparison.

For the Bimolecular Fluorescence Complementation assay (BiFC)

Agrobacterium was infiltrated into *N. benthamiana* as previously described (Jiao *et al.*, 2019 [DOI](#)). Fluorescence signals were observed after 3 days. The full description of BiFC is contained in Methods S9 and S10 of Supplemental Data 1.

For the observation of SINCS-like condensates

Agrobacterium was infiltrated into *N. benthamiana*. After 2 days, the leaves were treated in 1 mM SA solution for 24 h, and then fluorescence signals were observed. At least 20 image sets were obtained and analyzed. A full description of SINCS-like condensates observation is included in Methods S11 of Supplemental Data 1.

For growth of *Pst* DC3000/*avrRps4*

A low dose ($OD_{600} = 0.001$) of *Pst* DC3000/*avrRps4* was used for the infiltration experiments. After 3 days, the colony count was counted according to a previous description (Wang *et al.*, 2016 [DOI](#); Lei *et al.*, 2020 [DOI](#)). Full description of the growth of *Pst* DC3000/*avrRps4* is given in Methods S12 of Supplemental Data 1.

Free SA measurement

Free SA was extracted from 3-week-old *Arabidopsis* using a previously described method (Wang *et al.*, 2016 [DOI](#); Gong *et al.*, 2020 [DOI](#)). Free SA was measured by High-performance liquid chromatography (Shimadzu LC-6A, Japan). Detection conditions: 294 nm excitation wavelength, 426 nm emission wavelength.

Real-Time Quantitative PCR (RT-qPCR)

Total RNA was extracted from *Arabidopsis* (100 mg) using Trizol RNA reagent (Invitrogen, 10296-028, Waltham, MA, USA). RT-qPCR assays were performed as previously described (Zhang *et al.*, 2018 [DOI](#); Zhang *et al.*, 2022 [DOI](#)). All primers for RT-qPCR are listed individually in Table S5 of Supplemental Data 1. Full description of RT-qPCR is included in Methods S13 of Supplemental Data 1.

Trypan Blue Staining

The leaves of 3-week-old Col, amiRNA^{ATG6} # 1, amiRNA^{ATG6} # 2, *npr1*, *NPR1-GFP*, *ATG6-mCherry* and *ATG6-mCherry* × *NPR1-GFP* plants, located in the fifth and sixth positions, were infiltrated with *Pst* DC3000/*avrRps4*. After 3 days, the leaves were excised and subjected to a 1 min boiling step in trypan blue staining buffer (consisting of 10 g phenol, 10 mL glycerol, 10 mL lactic acid, 10 mL ddH₂O, and 10 mg trypan blue), followed by destaining 3 times at 37°C in 2.5 mg/mL chloral hydrate.

Statistical Analysis

All quantitative data in this study were presented as mean ± SD. The experimental data were analyzed by a two-tailed Student's *t*-test. Significance was assigned at *P* values < 0.05 or < 0.01.

Abbreviations:

artificial miRNA^{ATG6}(amiRNA^{ATG6})

Autophagy-related genes (ATGs)

Cycloheximide (CHX)

Gold nanoparticles (AuNPs)

Nonexpressor of pathogenesis-related genes 1 (NPR1)

Nuclear localization sequence (NLS)

P. syringae pv. *tomato* (*Pst*)

SA-induced NPR1 condensates (SINCs)

Salicylic acid (SA)

UBQ10::ATG6-mCherry × *35S::NPR1-GFP* (*ATG6-mCherry* × *NPR1-GFP*).

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Author contribution

B.-H.Z., J.Z. and W.-L.C. designed and supervised the research; B.-H.Z. and S.-Q.H. performed majority experiments; B.-H.Z., Y.-X.M. and H.C. synthesized gold nanoparticles (AuNPs); B.-H.Z. and S.-Q.H., Y.-Z.T. performed western blotting analysis; B.-H.Z., Y.Z and X.L performed confocal experiment; B.-H.Z. and S.-Q.H. analyzed the data; B.-H.Z. and S.-Q.H. prepared figures; B.-H.Z. and S.-Q.H. wrote the manuscript; J.Z. and W.-L.C. supervised and completed the writing; S.-Y.G. drew working model and edited language. J.Z. and W.-L.C. provided reagents/materials/analysis tools; J.Z. and W.-L.C. agrees to serve as the corresponding author responsible for contact and ensures communication.

Conflicts of interest

The author declares that there are no conflicts of interest.

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Data availability

The authors confirm that the data supporting the results of this study are available in the article and its supplementary materials.

List Supplemental Information

Supplemental Data 1

Supplemental Figure

Fig. S1 Physical interaction between NPRs and ATGs in yeast.

Fig. S2 Co-localization of NPR1-GFP and ATG6-mCherry in *N.benthamiana* under normal and SA treatment conditions.

Fig. S3 The nuclear localization of ATG6 in *Arabidopsis*. Fig. S4 Identification of *ATG6-mCherry* × *NPR1-GFP* plants.

Fig. S5 Subcellular fractionation of endogenous ATG6 in Col after 0.5 mM SA treatment for 0, 3, 6 and 20 h.

Fig S6. Confocal images of NPR1-GFP nuclear localization in 7-day-old seedlings of *NPR1-GFP* and *ATG6-mCherry* × *NPR1-GFP* under normal and 0.5 mM SA spray for 3 h.

Fig. S7 Overexpression of *ATG6* delayed dark-induced leaf senescence.

Fig. S8 Expression of *ICS1* under normal and SA treatment conditions.

Fig. S9 Expression of *PR1* and *PR5* in Col and *ATG6-mCherry* under normal and *Pst* DC3000/*avrRps4* treatment.

Fig. S10 The protein level of NPR1-GFP in *NPR1-GFP*/silencing *ATG6* and *NPR1-GFP*/Negative control.

Fig. S11 Expression of *NPR1* in Col and *ATG6-mCherry* under normal and *Pst* DC3000/*avrRps4* treatment.

Fig. S12 Partial co-localization of *ATG6-mCherry* and SINC-like condensates. Fig S13. *ATG6* improves the protein stability of *NPR1* in *N. benthamiana*.

Fig. S14 Structural analysis of acidic activation domains in *ATG6*.

Fig. S15 *ATG6* and *NPR1* cooperatively inhibit *Pst* DC3000/*avrRps4*-induced cell dead.

Fig. S16 *NPR1-GFP* degradation assay in *ATG6-mCherry* x *NPR1-GFP Arabidopsis*.

Fig. S17. Verification of *ATG6* antibody specificity.

Supplemental Table

Table S1. Plasmid in this study.

Table S2. Primers for vector construction.

Table S3. Plant Materials.

Table S4. Antibody Information.

Table S5. Primers of RT-qPCR.

Supplemental Results

Results S1. *NPR1* and its paralogues *NPR3/NPR4* physically interact with multiple *ATGs*.

Results S2. Overexpression of *ATG6* delays carbon starvation-induced leaf senescence, and *ATG6-GFP* and *ATG6-mCherry* fusion proteins are functional.

Supplement Methods

Methods S1. Plasmid construction Methods S2. *Arabidopsis* screening.

Methods S3. For *Pst* DC3000/*avrRps4* culture and infiltration.

Methods S4. Yeast two-hybrid assay.

Methods S5. Prokaryotic protein expression.

Methods S6. Nuclear and cytoplasmic separation of NPR1-GFP.

Methods S7. Protein degradation analysis.

Methods S8. Protein extraction and western blotting.

Methods S9. For the treatment of 3-week-old *N. benthamiana*.

Methods S10. For the bimolecular fluorescence complementation assay. Methods S11. For SINC-like condensates observation.

Methods S12. For the growth of *Pst* DC3000/*avrRps4*.

Methods S13. Real-Time Quantitative PCR (RT-qPCR).

Accession numbers Supplemental Moves

Supplemental Move 1. Localization of NPR1-GFP in *N. benthamiana* co-expressed NPR1-GFP and mCherry.

Supplemental Move 2. Localization of NPR1-GFP in *N. benthamiana* co-expressed NPR1-GFP and ATG6-mCherry.

Supplemental Data 2

Vector mapping used in this study.

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Reviewer #1 (Public Review):

The authors showed that autophagy-related genes are involved in plant immunity by regulating the protein level of the salicylic acid receptor, NPR1.

The experiments are carefully designed and the data is convincing. The authors did a good job of understanding the relationship between ATG6 and NPR1.

The authors have addressed most of my previous concerns.

<https://doi.org/10.7554/eLife.97206.2.sa2>

Reviewer #2 (Public Review):

The manuscript by Zhang et al. explores the effect of autophagy regulator ATG6 on NPR1-mediated immunity. The authors propose that ATG6 directly interacts with NPR1 in the nucleus to increase its stability and promote NPR1-dependent immune gene expression and pathogen resistance. This novel role of ATG6 is proposed to be independent of its role in autophagy in the cytoplasm. The authors demonstrate through biochemical analysis that ATG6 interacts with NPR1 in yeast and very weakly in vitro. They further demonstrate using

overexpression transgenic plants that in the presence of ATG6-mcherry the stability of NPR1-GFP and its nuclear pool is increased.

Comments on revised version:

The authors demonstrate the correlation between overexpression of atg6 and higher stability and activity of npr1. They claim a novel activity of atg6 in the nucleus. Overall, the experimental scope of the study is solid, however, the over-interpretation of the results substantially reduces the significance and value of this study for the target plant immunity readership.

<https://doi.org/10.7554/eLife.97206.2.sa1>

Author response:

The following is the authors' response to the original reviews.

eLife assessment

The study reports on a previously unrecognized function of ATG6 in plant immunity. The work is valuable because it proposes a direct interaction between ATG6 and a well-studied salicylic acid receptor protein, NPR1, which may interest researchers investigating plant immunity regulation. While the data presented are compelling, more information regarding the specificity of ATG6's role would improve the overall impact of the study, especially with an eye towards consistency with prior work.

We also genuinely thank the editor and reviewers for the constructive and helpful suggestions and comments. These comments have greatly improved the quality and thoroughness of our manuscript. We have carefully studied these comments and have made the appropriate changes as far as possible. Additionally, some minor errors were also corrected during the revision process. New text is shown in blue in the revised manuscript. Our responses to the reviewer's comments are provided below each respective comment.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

The authors showed that autophagy-related genes are involved in plant immunity by regulating the protein level of the salicylic acid receptor, NPR1.

Strengths:

The experiments are carefully designed and the data is convincing. The authors did a good job of understanding the relationship between ATG6 and NPR1.

Thank you very much for recognizing our research.

Weaknesses:

- The authors can do a few additional experiments to test the role of ATG6 in plant immunity.

I recommend the authors to test the interaction between ATGs and other NPR1 homologs (such as NPR2).

Thanks to your valuable feedback, it was discovered that the Arabidopsis NPRs family comprises six members: NPR1, NPR2, NPR3, NPR4, NPR5/PETIOLE 1 (BOP1), and NPR6/BOP2. NPR3/4 function in tandem as negative regulators to modulate SA signaling and plant immune responses (Ding et al., 2018). Similar to NPR1, NPR2 acts as a positive regulator of SA

signaling (Castello et al., 2018). NPR5/BOP1 and NPR6/BOP2 primarily participate in the regulation of plant growth and development (McKim et al., 2008). This study specifically investigates the correlation between ATG6 and NPRs in plant resistance to pathogenic bacteria. Consequently, we experimentally confirmed the interaction between ATG6 and NPR1, NPR3, and NPR4 (Fig. 1 and Fig. S1 in the revised manuscript). It would be intriguing to further explore the interactions between ATG6 and other NPRs in the context of regulating plant growth and development in future research endeavors.

-The concentration of SA used in the experiment (0.5-1 mM) seems pretty high. Does a lower concentration of SA induce ATG6 accumulation in the nucleus?

Thank you for pointing this out. The NPR1 protein is known to be unstable and prone to degradation through the 26S proteasome pathway (Spoel et al., 2009; Saleh et al., 2015). Consequently, to investigate the function of NPR1, many scientists and research groups typically employ higher concentrations of SA (e.g., 0.5 mM, 1 mM, or even 5 mM) to elucidate its role (Spoel et al., 2009; Fu et al., 2012; Lee et al., 2015; Saleh et al., 2015; Skelly et al., 2019; Zavaliev et al., 2020; Chen et al., 2021a). In our study, we observed an interaction between ATG6 and NPR1. To enhance the detection of the NPR1 protein, we standardized the SA concentration (Arabidopsis was treated with 0.5 mM SA; Tobacco was treated with 1 mM SA) used in our experiments. Subsequently, we analyzed the nuclear accumulation ATG6 or NPR1 using a relatively high SA concentration (Arabidopsis was treated with 0.5 mM SA; Tobacco was treated with 1 mM SA), consistent with concentrations used in previous studies (Spoel et al., 2009; Lee et al., 2015; Saleh et al., 2015; Skelly et al., 2019; Zavaliev et al., 2020; Chen et al., 2021a).

-Does the silencing of ATG6 affect the cell death (or HR) triggered by AvrRPS4?

Thank you for pointing this out. In this study, we examined changes in Pst DC3000/avrRps4-induced cell death in Col, amiRNAATG6 # 1, amiRNAATG6 # 2, npr1, NPR1-GFP, ATG6-mCherry and ATG6-mCherry × NPR1-GFP plants. The results of Taipan blue staining showed that Pst DC3000/avrRps4-induced cell death in npr1, amiRNAATG6 # 1 and amiRNAATG6 # 2 was significantly higher compared to Col (Fig. S15 in the revised manuscript). Conversely, Pst DC3000/avrRps4-induced cell death in ATG6-mCherry, NPR1-GFP and ATG6-mCherry × NPR1-GFP was significantly lower compared to Col. Notably, Pst DC3000/avrRps4-induced cell death in ATG6-mCherry × NPR1-GFP was significantly lower compared ATG6-mCherry and NPR1-GFP (Fig. S15 in the revised manuscript). These results suggest that ATG6 and NPR1 cooperatively inhibit Pst DC3000/avrRps4-induced cell death. The relevant description can be found in lines 394-404 of the revised manuscript.

-SA and NPR1 are also required for immunity and are activated by other NLRs (such as RPS2 and RPM1). Is ATG6 also involved in immunity activated by these NLRs?

Thank you for your valuable comments. The most notable event in the NLR-mediated ETI immune response is the induction of hypersensitive response-programmed cell death (HR-PCD) (Jones and Dangl, 2006; Yuan et al., 2021). SA plays a dual role in the ETI response. On one hand, the accumulation of SA during the R gene-mediated ETI defense response is directly linked to the onset of HR-PCD (Nawrath and Metraux, 1999). SA and NPR1 can enhance the ETI response by regulating the expression of downstream target genes (Falk et al., 1999; Feys et al., 2001; Ding et al., 2018; Liu et al., 2020). On the other hand, the activation of SA signaling can have a negative regulatory effect on HR-PCD during the ETI response. High levels of SA have been shown to significantly inhibit HR-PCD triggered by the avrRpt2 effector (Rate and Greenberg, 2001; Devadas and Raina, 2002; Jurkowski et al., 2004). Rate et al. discovered that the inhibition of HR-PCD by SA relies on NPR1 (Rate and Greenberg, 2001).

Arabidopsis AtATG6 or its homologs in other species (such as NbBECLIN1, TaATG6s, etc.) have been identified as positive regulators in plant immunity, playing a crucial role in inhibiting cell death and preventing invasion by pathogenic microorganisms (Liu et al., 2005; Patel and Dinesh-Kumar, 2008; Yue et al., 2015). Patel et al. demonstrated that, akin to autophagy-deficient mutants previously documented, AtATG6 antisense (AtATG6-AS) plants treated with Pst DC3000/avrRpm1 exhibited diffuse cell death, indicating the necessity of ATG6 in restricting cell death (Patel and Dinesh-Kumar, 2008). In tobacco, deficiencies in BECLIN 1 result in the onset of diffuse HR-PCD, underscoring the essential role of BECLIN 1 in limiting HR-PCD (Liu et al., 2005). Despite the genetic evidence supporting the critical function of ATG6 in plant immunity, the precise molecular mechanisms through which ATG6 impedes the invasion of pathogenic microorganisms remain elusive.

In our study, we uncovered that ATG6 interacts with NPR1 to hinder pathogen invasion and inhibit the initiation of cell death. In animals, members of the NLR family have been observed to interact with the autophagy-related protein LC3 to inhibit the survival of pathogen (Zhang et al., 2019). Similar mechanisms may exist in plants. However, it remains to be explored whether NLR directly induces the activation of ATG6 through interaction or the relationship between NPR1-ATG6 interactions and NLR-mediated plant immunity, necessitating further investigation.

Reviewer #2 (Public Review):

Summary:

The manuscript by Zhang et al. explores the effect of autophagy regulator ATG6 on NPR1-mediated immunity. The authors propose that ATG6 directly interacts with NPR1 in the nucleus to increase its stability and promote NPR1-dependent immune gene expression and pathogen resistance. This novel role of ATG6 is proposed to be independent of its role in autophagy in the cytoplasm. The authors demonstrate through biochemical analysis that ATG6 interacts with NPR1 in yeast and very weakly in vitro. They further demonstrate using overexpression transgenic plants that in the presence of ATG6-mcherry the stability of NPR1-GFP and its nuclear pool is increased.

However, the overall conclusions of the study are not well supported experimentally. The significance of the findings is low because of their mostly correlational nature, and lack of consistency with earlier reports on the same protein.

Thank you for your valuable and constructive suggestions. In this article, we unveil a novel relationship in which ATG6 positively regulates NPR1 in plant immunity (Fig. 8 in the revised manuscript). ATG6 interacts with NPR1 to synergistically enhance plant resistance by regulating NPR1 protein levels, stability, nuclear accumulation, and formation of SINCs-like condensates. This may be of interest to researchers studying the regulation of plant immunity. While there may be minor flaws in our current study, the significance of these findings cannot be overstated, as they have the potential to redirect scientific attention towards uncovering novel functions for autophagy genes.

Based on the integrity and quality of the data as well as the depth of analysis, it is not yet clear if ATG6 is a specific regulator of NPR1 or if it is affecting NPR1's stability indirectly, through inducing an elevation of SA levels in plants. As such, the current study demonstrates a correlation between overexpression of ATG6, SA accumulation, and NPR1 stability, however, whether and how these components work together is not yet demonstrated.

Thanks to your valuable feedback. Although as the reviewer said there may be some flaws in our data from the current results, scientific research is an ongoing process and I am

confident that future studies will be even better. From the results given to us at the moment at least this study reports a previously undiscovered function of ATG6 in plant immunity. We propose a direct interaction between ATG6 and a well-studied salicylic acid receptor protein, NPR1. We unveil a novel relationship in which ATG6 positively regulates NPR1 in plant immunity (Fig. 8 in the revised manuscript). ATG6 interacts with NPR1 to synergistically enhance plant resistance by regulating NPR1 protein levels, stability, nuclear accumulation, and formation of SINC-like condensates. This may be of interest to researchers studying the regulation of plant immunity.

Based on the provided biochemical data, it is not yet clear if the ATG6 functions specifically through NPR1 or through its paralogs NPR3 and NPR4, which are negative regulators of immunity. It is quite possible that interaction with NPR1 (or any NPR) is not the major regulatory step in the activity of ATG6 in plant immunity. The effect of ATG6 on NPR1 could well be indirect, through a change in the SA level and redox environment of the cell during the immune response. Both SA level and redox state of the cell were reported to induce accumulation of NPR1 in the nucleus and increase in stability.

Thanks to your valuable feedback. In this study, we validated the interaction between ATG6 and NPR1 through various approaches and identified the key regions mediating their interaction. Our findings indicate that ATG6 interacts with NPR1 to synergistically enhance plant resistance by regulating NPR1 protein levels, stability, nuclear accumulation, and the formation of SINC-like condensates. These results clearly demonstrate the involvement of ATG6 in the regulation of NPR1. Furthermore, we also found that ATG6 interacts with NPR3/4 (Fig. S1 in the revised manuscript). This is particularly relevant given that NPR3 and NPR4 have been shown to act as adaptors for the ubiquitin E3 ligase Cullin 3 (CUL3) to regulate the degradation of NPR1. Therefore, whether ATG6 regulates NPR1 through its interactions with NPR3/4 is an intriguing question worth exploring in future studies. We appreciate the reviewer's concerns and are committed to addressing them in our future research to further elucidate the complex regulatory mechanisms involving ATG6, NPR1, and other key players in plant immunity.

Another major issue is the poor quality of the subcellular analyses. In contradiction to previous studies, ATG6 in this study is not localized to autophagosome puncta, which suggests that the soluble localization pattern presented here does not reflect the true localization of ATG6. Even if the authors propose a novel, non-canonical nuclear localization for ATG6, they still should have detected the canonical autophagy-like localization of this protein.

Thanks to your valuable feedback. We conducted predictions at NLS Mapper (https://nls-mapper.iab.keio.ac.jp/cgi-bin/NLS_Mapper_form.cgi) and identified two bipartite NLSs in ATG6, with the sequences "MRKEEIPDKSRTIPIDPNLPKWVCQNCHHS" and "DPNLPKWVCQNCHHS LTIVGVDSYAGKFFNDP". To further elucidate the nuclear localization of ATG6, we introduced *Agrobacterium tumefaciens* carrying ATG6-GFP into *nls-mCherry* tobacco leaves through transient transformation. Subsequently, we observed the localization of ATG6-GFP, along with the canonical autophagy-like patterns. Our findings revealed fluorescence signals of ATG6-GFP in both the cytoplasm and nuclei (Figure 2b). The nuclear-localized ATG6-GFP overlapping with the nuclear-localized marker, *nls-mCherry* (indicated by white arrows). Additionally, we observed punctate patterns indicative of canonical autophagy-like localization of ATG6-GFP fluorescence signals (indicated by red circles). Based on these results, we are more confident about the authenticity of ATG6's nuclear localization. The revised manuscript includes clearer images to support our observations.

Recommendations for the Authors:

Reviewer #2 (Recommendations For The Authors):

The duration and concentration of SA treatments are quite variable between experiments which makes comparisons difficult.

Thank you for pointing this out. The NPR1 protein is known to be unstable and prone to degradation through the 26S proteasome pathway (Spoel et al., 2009; Saleh et al., 2015). Consequently, to investigate the function of NPR1, many scientists and research groups typically employ higher concentrations of SA (e.g., 0.5 mM, 1 mM, or even 5 mM) to elucidate its role (Spoel et al., 2009; Fu et al., 2012; Lee et al., 2015; Saleh et al., 2015; Skelly et al., 2019; Zavaliev et al., 2020; Chen et al., 2021a). In our study, we observed an interaction between ATG6 and NPR1. To enhance the detection of the NPR1 protein, we standardized the SA concentration used in our experiments. In this study, for the treatment of Arabidopsis, we followed the protocols outlined in Saleh et al. and Spoel et al., utilizing 0.5 mM SA (Spoel et al., 2009; Saleh et al., 2015). For tobacco treatment, we adopted the methodology described in the study by Zavaliev et al., administering 1 mM SA (Zavaliev et al., 2020).

The methods section does not explain some of the essential experimental conditions and reagents used in the study.

Thank you for pointing this out. Due to word limitations we have placed the detailed experimental methods and reagents in Supplemental Data 1. In Supplemental Data 1, we provide a comprehensive overview of the experimental flow and conditions employed in our study.

Lines 62-63: the C-terminal domain of all NPRs has a name (already defined as SA-binding domain (SBD)). Also, it would be worth referring to the structure of NPR1 (Kumar et al 2022, Nat) as the source of information about its domains.

Thank you for pointing this out, we have changed this description in the revised manuscript (lines 62-63).

Lines 66-69: NPR1 doesn't form monomers. A recent study showed that the basic functional unit of NPR1 is a dimer (Kumar et al 2022, Nat).

Thank you for pointing this out. In the revised manuscript (line 67) " monomers " has been changed to "dimer".

Lines 89-95 and elsewhere: the term "invasion" has a very specific meaning and it doesn't necessarily refer to disease. A pathogen can invade the plant but cause no disease (e.g. ETI). Most plant genetic immune mechanisms act after pathogen invasion, not before it. Those cited works reported the disease resistance, not the invasion resistance.

Thank you for pointing this out. We've changed the incorrect description in the revised manuscript (line 91).

Lines 113-119: the truncation at the aa328 includes half of the ANK domain (repeats 1 and 2), not just BTB. The C-terminal truncation variant contains the other half (repeats 3 and 4) of the ANK domain, not the entire ANK domain. It also contains the SBD, not just the NLS. So, this kind of analysis cannot determine the role of ANK domain in the interaction, nor it can conclusively determine if the interaction is through SBD. The interaction should be tested with the SBD domain only in order to make this conclusion.

Thank you for pointing this out, we have removed the inappropriate description and made the appropriate changes in the revised manuscript (lines 114 and 115).

In Figure S1, the equally strong interaction of atg6 is found for NPR3/NPR4. Does that mean that atg6 functions also through these other NPRs? What's the significance of these data compared to NPR1-ATG6 interaction? This is especially important, because both NPR3 and NPR4 are predominantly nuclear proteins, and they are unlikely to significantly overlap with autophagy components in the cytoplasm.

NPR1 and its paralogues NPR3/NPR4, which frequently interact with other proteins to regulate plant immune responses (Backer et al., 2019; Chen et al., 2019). To identify ATGs that interact with NPRs, we performed yeast two-hybrid (Y2H) screens using NPRs as bait. Interestingly, ATG6 interacted with NPR1, NPR3 and NPR4, respectively, and different concentrations of SA treatment did not significantly affect their interaction (Fig. S1a). NPR1 is an important positive regulator of the plant immune response (Chen et al., 2021b). In *Arabidopsis* and *N. benthamiana*, ATG6 or its homologues was reported to act as a positive regulator to enhance plant disease resistance to *P. syringae* pv. tomato (Pst) DC3000 and Pst DC3000/avrRpm1 bacteria (Patel and Dinesh-Kumar, 2008), *N. benthamiana* mosaic virus (TMV) (Liu et al., 2005). Therefore, in this study we focused on investigating the biological significance of the interaction between ATG6 and NPR1. Whether the interaction between ATG6 and NPR3/4 also has an effect on plant immunity is a question that remains to be explored in future studies.

In Figure 1c and elsewhere: why not use the anti-mCherry antibody to detect atg6-mcherry? Are we seeing the correct protein band of atg6-mcherry? Also, it is not clear what antibodies they used throughout the study: the sources and specificities of antibodies are not provided.

Thank you for pointing this out. We initially synthesized the ATG6 antibody (anti-ATG6, 1:200, peptide, C-KEKKKIEEEERK, Abmart) in order to detect the endogenous ATG6 protein, and we also tested the specificity and potency of the ATG6 antibody (results are shown in Fig. S17). Additionally, in order to determine the location of the ATG6-mCherry bands, we also detected ATG6-mCherry in ATG6-mCherry *Arabidopsis* using the ATG6 antibody, and we also used Col as a control (results are shown in Fig. S4). These results show that our synthesized ATG6 antibody can effectively and clearly immunize to both ATG6 and ATG6-mCherry. Therefore, in this study, we used the ATG6 antibody to analyze both ATG6-mCherry and endogenous ATG6. Detailed antibody information is presented in Supplementary Data 1, table S4

In Figures 1d, 2a, and 2b, the subcellular localization pattern of atg6 contradicts what was published before (Fujiki et al 2007, Plant Phys; Liu et al 2018, FPIS; Xu et al 2017, Autophagy; Li et al 2018, Nat. Comm.). As an autophagy protein, atg6 was shown to localize to cytoplasmic puncta (autophagosomes), like atg8. No nuclear localization was found in those studies. The lack of puncta and the strong nuclear accumulation are signs that the localization of atg6 reported here has to be interpreted with caution. With the data provided, I am not convinced yet that we are looking at the correct ATG6 subcellular localization. Even if the authors propose a novel, non-canonical localization for atg6, they still should have detected the canonical autophagy-like localization of this protein.

Thanks to your valuable feedback. To further elucidate the nuclear localization of ATG6, we introduced *Agrobacterium tumefaciens* carrying ATG6-GFP into nls-mCherry tobacco leaves through transient transformation. Subsequently, we observed the localization of ATG6-GFP, along with the canonical autophagy-like patterns. Our findings revealed fluorescence signals of ATG6-GFP in both the cytoplasm and nuclei (Figure 2b). The nuclear-localized ATG6-GFP

overlapping with the nuclear-localized marker, nls-mCherry (indicated by white arrows). Additionally, we observed punctate patterns indicative of canonical autophagy-like localization of ATG6-GFP fluorescence signals (indicated by red circles). Based on these results, we are more confident about the authenticity of ATG6's nuclear localization. The revised manuscript includes clearer images to support our observations.

It would make more sense to include the BiFC data (fig. S2) in the main figure, instead of the co-localization (fig. 1d) which cannot serve as evidence for interaction.

Thank you for the feedback. We accept your suggestion. In Fig.1, we have replaced the co-localization image with a BiFC (Bimolecular Fluorescence Complementation) image to better illustrate the interaction.

In Figure S2, the bifc signals have to be quantified to qualify as evidence for interaction. also, a subcellular marker has to be used (e.g. nuclear mcherry). From the current poor-quality images, one cannot determine where in the cell the presumed interaction takes place, nucleus or cytoplasm, or both. Also, no puncta are seen in these images.

Thank you for pointing this out. Despite the lack of clarity in the images we provided, our BiFC results unequivocally demonstrate the interaction between ATG6 and NPR1 in both the cytoplasm and nucleus. Notably, as the reviewer pointed out, punctate signals were not observed in our images. This lack of punctate signals is consistent with previous studies (Figure 2) that have also shown BiFC results between autophagy-associated proteins ATG8s and their interacting partners. For instance, Fig 1G (Marshall et al. 2019, Cell), Fig 2F (Marshall et al. 2019, Cell), Fig 4B (Macharia et al. 2019, BMC Plant Biology), and Fig 3 (Zhou et al. 2018, Autophagy) all did not exhibit punctate signals, aligning closely with our findings.

In Figure S3a, the nuclear localization is shown for stomata. It is known that stomata are especially strong expressors of the transgenes, and localization there could be an artefact of overaccumulation of the fusion protein. Also, why do they present the localization of atg6-gfp, if the analysis and the cross were made with atg6-mcherry?

Thank you for pointing this out. In our previous experiments, we observed the localization of ATG6 in the nucleus of *Arabidopsis thaliana* plants overexpressing ATG6-GFP (Fig. S3a). To clearly visualize the location of the nucleus, we used the cytosolic DAPI dye, which readily stained the nuclei of the stomatal guard cells. This allowed us to easily identify the nuclear regions for our observations. Additionally, in Fig. 2a and Fig.S3b, we detected the fluorescence signal of ATG6-mCherry within the nucleus, further confirming the nuclear localization of ATG6. Moreover, the nuclear and cytoplasmic fractions were separated. Under SA treatment, ATG6-mCherry and ATG6-GFP were detected in the cytoplasmic and nuclear fractions in *N. benthamiana* (Fig. 2c and d). Similarly, ATG6 was also detected in the nuclear fraction of UBQ10::ATG6-GFP and UBQ10::ATG6-mCherry overexpressing plants (Fig. 2e and f).

In Figure S3b, the images are low resolution and of poor quality. Why atg6-mcherry is expressed in a single cell if these are transgenic plants? The nuclear co-localization with npr1-gfp has to be shown more clearly with high res. images and also be quantified, because the expression of atg6-mcherry is not as uniform as npr1-gfp.

Thank you for pointing this out. Contrary to the reviewer's assertion, the ATG6-mCherry fluorescence signal depicted in Figure S3b was not exclusive to a single cell. In fact, this fluorescence was also evident in other cells, albeit with relatively weaker intensity. This disparity in fluorescence intensity may be attributed to the irregularities in leaf structure at the time of image capture using the microscope. To bolster our conclusion, we further

examined the fluorescence signals in the cells of the root elongation zone in ATG6-mCherry x NPR1-GFP, as depicted in the figure below. Our observations revealed that the fluorescence signals of ATG6-mCherry exhibited uniform distribution, with detection in both the cytoplasm and nucleus. We have replaced the original unclear image with a high-quality image.

Lines 138-143: In fig. S3d, it would make more sense to show the WB on the hybrid npr1-gfp/atg6-mcherry plants with both anti-gfp and anti-mcherry antibodies to detect the free mcherry/gfp. Since the analysis of the level of free FP is done, then why didn't they test the free mcherry levels in Figure S4a? This would be more important than testing the free GFP in ATG6-GFP plants, because the imaging of atg6-mcherry was done in the hybrid plants (fig. S3b).

Thank you for pointing this out. We initially synthesized the ATG6 antibody (anti-ATG6, 1:200, peptide, C-KEKKKIEEEERK, Abmart) in order to detect the endogenous ATG6 protein, and we also tested the specificity and potency of the ATG6 antibody (results are shown in Fig. S17). Additionally, in order to determine the location of the ATG6-mCherry bands, we also detected ATG6-mCherry in ATG6-mCherry Arabidopsis using the ATG6 antibody, and we also used Col as a control (results are shown in Fig. S4). These results show that our synthesized ATG6 antibody can effectively and clearly immunize to both ATG6 and ATG6-mCherry. Therefore, in this study, we used the ATG6 antibody to analyze both ATG6-mCherry and endogenous ATG6. Detailed antibody information is presented in Supplementary Data 1, table S4. In the previous experiments, we procured the mCherry antibody (mCherry-Tag Monoclonal Antibody(6B3), BD-PM2113, China) to immunolabel ATG6-mCherry. However, we encountered challenges with the potency of this mCherry antibody, and considering our budget constraints, as well as the availability of our self-synthesized ATG6 antibody, we chose not to pursue the purchase of another antibody from a different company for the continuation of the Western Blot experiment.

In Figure 2c, there's no atg6-mcherry detected at time 0, in either cytoplasm or nucleus, yet the microscope images in panel a show strong accumulation in both compartments.

Thank you for pointing this out. Previous studies ATG6 can also be degraded via the 26S proteasome pathway (Qi et al., 2017). We speculate that this phenomenon might be attributed to the rapid turnover of ATG6 at time 0.

Lines 156-160: this statement is unsupported by the data. In fig. S5, the bands for native atg6 in the nuclear fraction are extremely weak, and they do not show the reverse pattern of change along the time points compared to the cytoplasmic fraction, which would indicate that the nuclear fraction is complementary to the cytoplasmic pool of the protein. The result more likely suggests that the majority of the ATG6 is in the cytoplasm, and that the weak bands detected in the nucleus are either background signal, or a contamination from the cytoplasmic pool. At this low protein level or poor immuno-detection the background signal is inevitable due to overexposure. Even though the actin marker is not detected in the nuclear fraction, it doesn't necessarily mean that there's no contamination from the cytoplasm in the nuclear fraction. The actin is just too abundant and can be detected at lower exposure.

Thank you for pointing this out. In Fig. S5, we detected the subcellular localization of endogenous ATG6, although the image quality was somewhat low. Nevertheless, the cytosolic and nuclear localization of ATG6 could be clearly observed. In addition to this, we also verified the cytosolic and nuclear localization of ATG6 in Arabidopsis using confocal fluorescence microscopy and nucleoplasmic separation experiments. Actin and H3 were used as cytoplasmic and nucleus internal reference, respectively. (Fig. 2e and f). Furthermore, we observed the cytosolic and nuclear localization of ATG6 when we expressed ATG6-GFP or

ATG6-mCherry in tobacco leaves through cis-transfection experiments (Fig. 2a-d). These results are consistent with the prediction of the subcellular location of ATG6 in the Arabidopsis subcellular database (<https://suba.live/>) (Fig. S3c). The reviewer's feedback has been valuable in helping us present these findings more clearly. We acknowledge the limitations in the image quality for the endogenous ATG6 localization, but we believe the combination of multiple experimental approaches, including the use of fluorescent protein fusions, provides robust evidence for the cytosolic localization of ATG6 in plant cells. Moving forward, we will continue to investigate the significance of ATG6's subcellular distribution and its potential dual roles in both the nucleus and the cytosol, particularly in the context of its interaction with the key immune regulator NPR1. We appreciate the reviewer's constructive comments, as they will help us strengthen the presentation and interpretation of our findings.

In Figure 3a the images are of too low resolution to see the co-localization. The focal planes of the top and bottom panels are quite different: the top is focused on stomata, the bottom - on pavement cells. So, the number of the NPR1-GFP nuclei between these two focal planes is dramatically different. Also, it looks like the atg6-mcherry in these plants are predominantly in the cytoplasm, not the nucleus as the authors claim. A higher resolution and higher quality of images are required to determine this.

Thank you for pointing this out. To ensure the clarity and accuracy of our confocal images, we have supplied a clearer image as supplementary evidence. The Bright images distinctly show that both sets of images are in the same plane of focus. Furthermore, in the figure (third one in the fourth column), the nucleus localization of ATG6-mCherry is clearly visible, and that ATG6-mCherry is co-localized with NPR1-GFP in the nucleus, as indicated by the white arrow.

In Figure 3b, it is not indicated what exactly was measured and in what condition, mock or SA. If these are numbers of nuclei, then it should be indicated what size of the area was sampled, not just "section", and both mock and SA should be included in the measurements. Also, how many independent images have been sampled? what does the error bar represent? What does "normal" mean? Shouldn't this be a mock treatment?

Thank you for pointing out this. The term "Normal" in this context refers to mock treatment, and we have revised the description for clarity. In Figure 3b, the graph illustrates the count of nuclear localizations of NPR1-GFP in ATG6-mCherry $\times$ NPR1-GFP and NPR1-GFP Arabidopsis plants following SA treatment. Statistical data were obtained from three independent experiments, each comprising five individual images, resulting in a total of 15 images analyzed for this comparison. Detailed descriptions were also added to the revised manuscript (Lines 568-570, 800-804).

Lines 167-168: the proposed increase of NPR1-GFP in the nucleus could be simply due to a higher accumulation of SA in the hybrid plants, not because of the direct interaction of atg6.

Thank you for pointing out this. Our results confirmed that ATG6 overexpression significantly increased nuclear accumulation of NPR1 (Fig. 3). Notably, the ratio (nucleus NPR1/total NPR1) in ATG6-mCherry $\times$ NPR1-GFP was not significantly different from that in NPR1-GFP, and there is a similar phenomenon in *N. benthamiana* (Fig. 3c-f). These results suggested that the increased nuclear accumulation of NPR1 by ATG6 might result from higher levels and more stable NPR1, rather than the enhanced nuclear translocation of NPR1 facilitated by ATG6. Furthermore, we found that under SA treatment, the protein levels of NPR1 were significantly higher in the ATG6-mCherry $\times$ NPR1-GFP line compared to the NPR1-GFP line (Fig. 5a). Notably, even in the absence of differences in SA levels between the two lines, we

observed that ATG6 could delay the degradation of NPR1 under normal conditions (Fig. 6). These findings suggest that ATG6 employs both SA-dependent and SA-independent mechanisms to maintain the stability of the key immune regulator NPR1. In summary, we therefore suggest that the increased nuclear accumulation in NPR1 cells is a dual effect of SA and ATG6.

Lines 202-204: "Increased nuclear accumulation" implies increased translocation. However, they found that the ratio of NPR1-GFP does not change (Figure 3), so the reason for higher nuclear accumulation is not translocation, but abundance.

Thank you for pointing out this. Our results confirmed that ATG6 overexpression significantly increased nuclear accumulation of NPR1 (Fig. 3). ATG6 also increases NPR1 protein levels and improves NPR1 stability (Fig. 5 and 6). Therefore, we consider that the increased nuclear accumulation of NPR1 in ATG6-mCherry x NPR1-GFP plants might result from higher levels and more stable NPR1 rather than the enhanced nuclear translocation of NPR1 facilitated by ATG6. To verify this possibility, we determined the ratio of NPR1-GFP in the nuclear localization versus total NPR1-GFP. Notably, the ratio (nucleus NPR1/total NPR1) in ATG6-mCherry x NPR1-GFP was not significantly different from that in NPR1-GFP, and there is a similar phenomenon in *N. benthamiana* (Fig. 3c-f). These results suggested that the increased nuclear accumulation of NPR1 by ATG6 might result from higher levels and more stable NPR1, rather than the enhanced nuclear translocation of NPR1 facilitated by ATG6. Further we analyzed whether ATG6 affects NPR1 protein levels and protein stability. Our results show that ATG6 increases NPR1 protein levels under SA treatment and ATG6 maintains the protein stability of NPR1 (Fig. 5 and 6). These results suggested that the increased nuclear accumulation of NPR1 by ATG6 result from higher levels and more stable NPR1. The corresponding description is shown in revised manuscript (lines 338~352).

Lines 204-205: the co-localization in Figure 1d cannot be interpreted as interaction.

Thank you for the feedback. We have replaced the co-localization image with a BiFC (Bimolecular Fluorescence Complementation) image to better illustrate the interaction in Fig 1d.

What age of plants were used for the analysis in Figures 4 and S7? The age of the plant might significantly affect the free SA levels under control conditions.

Thank you for the feedback. In Figures 4 and S7, 3-week-old plants were used to determine salicylic acid (SA) levels and the expression of target genes. Figures 4 and S7 figure notes provide detailed descriptions (lines 818-819).

In Figure 5a they treat with SA, but the analysis in Figure S10 is done with the pathogen, so how can these data be correlated?

Thank you for pointing out this. Previous studies have demonstrated that pathogen infestation rapidly increases the salicylic acid (SA) content in plants, and the elevated SA then activates plant immune responses. Therefore, both pathogen treatment and direct SA treatment can activate SA-dependent plant immune responses. The NPR1 protein is known for its instability. In Figure 5a, we utilized a 0.5 mM SA treatment to assess the changes in NPR1 protein levels, as the impact of SA treatment is more immediate and pronounced.

Lines 241-242: In Figure 5b, it is not clear why there's no detection of NPR1-GFP and atg6-mcherry at time 0?? The levels of proteins in the transient assay are sufficiently high for detection by WB.

Thank you for pointing this out. The NPR1 protein is known to be unstable and prone to degradation through the 26S proteasome pathway (Spoel et al., 2009; Saleh et al., 2015). In addition, previous studies ATG6 can also be degraded via the 26s proteasome pathway (Qi et al., 2017). We speculate that this phenomenon might be attributed to the rapid turnover of NPR1 and ATG6 at time 0.

In Figures 5c-d, the quality of these images is very poor, and they do not clearly show the signs. What structure was exactly measured in these images? There are so many fluorescent bodies there, that it is not clear what are we looking at. Also, it is not clear why they did not show the mcherry channel? It would be important to see if the bodies in SA-treated plants show co-localization with atg6-mcherry autophagosomes (if these exist at all).

Thank you for pointing this out. Interestingly, similar to previous reports (Zavaliev et al., 2020), SA promoted the translocation of NPR1 into the nucleus, but still a significant amount of NPR1 was present in the cytoplasm (Fig. 3c and e). Previous studies have shown that SA increased NPR1 protein levels and facilitated the formation of SINCs in the cytoplasm, which are known to promote cell survival (Zavaliev et al., 2020). We therefore observed the fluorescence signal of SINCs-like condensates in the cytoplasm of tobacco leaves. After 1mM SA treatment, more SINCs-like condensates fluorescence were observed in *N. benthamiana* co-transformed with ATG6-mCherry + NPR1-GFP compared to mCherry + NPR1-GFP (Fig. 5c-d and Supplemental movie 1-2). We have a clearer demonstration in the supplemental video movie 1-2. Additionally, we observed that SINCs-like condensates signaling partial co-localized with certain ATG6-mCherry autophagosomes fluorescence signals.

Lines 245-247: so, is it atg6 or SA that increases the NPR1 levels? If this is due to SA, then the whole study doesn't have novelty, because we already know from previous works that SA increases the stability of npr1.

Thank you for pointing this out. Indeed, previous studies have shown that salicylic acid (SA) increases NPR1 levels and protein stability (Spoel et al., 2009; Saleh et al., 2015). In our experiments, we found that under SA treatment, the protein levels of NPR1 were significantly higher in the ATG6-mCherry × NPR1-GFP line compared to the NPR1-GFP line (Fig. 5a). Additionally, free SA levels were also significantly elevated in the ATG6-mCherry × NPR1-GFP line under pathogen challenge (Pst DC3000/avrRps4), but not under normal conditions (Fig. 4a). Furthermore, even in the absence of differences in SA levels between the two lines, we observed that ATG6 could delay the degradation of NPR1 under normal conditions (Fig. 6). These findings represent one of our new discoveries. These findings suggest that ATG6 employs both SA-dependent and SA-independent mechanisms to maintain the stability of the key immune regulator NPR1.

Lines 313-316: npr1 and atg6 can function independently from each other, so the term "jointly" is misleading. Based on the overall data provided in this manuscript it cannot be concluded that the two proteins work in one complex to control plant immunity.

Thank you for pointing this out. In the revised manuscript "jointly" has been changed to "cooperatively".

Lines 369-374: this speculation is beyond the main hypothesis claiming that atg6 functions through npr1. If atg6 can activate the transcription alone, then what is the significance of its activation of npr1? How can one distinguish between the two?

Thank you for pointing this out. Transcription activation by transcription factors typically requires at least two conserved structural domains: a transcription activation domain and a DNA-binding domain. However, ATG6 does not possess these two typical conserved structural domains found in canonical transcription factors. Given this structural context, it is unlikely that ATG6 would be able to directly activate transcription on its own. The lack of the canonical transcription factor domains in ATG6 suggests that it may not be able to function as a direct transcriptional activator. Previous studies have shown that acidic activation domains (AADs) in transcriptional activators (such as Gal4, Gcn4 and VP16) play important roles in activating downstream target genes. Acidic amino acids and hydrophobic residues are the key structural elements of AAD (Pennica et al., 1984; Cress and Triezenberg, 1991; Van Hoy et al., 1993). Chen et al. found that EDS1 contains two ADD domains and confirmed that EDS1 is a transcriptional activator with AAD (Chen et al., 2021a). Here, we also have similar results that ATG6 overexpression significantly enhanced the expression of PR1 and PR5 (Fig. 4b-c and S9), and that the ADD domain containing acidic and hydrophobic amino acids is also found in ATG6 (148-295 AA) (Fig. S14). We speculate that ATG6 might act as a transcriptional coactivator to activate PRs expression synergistically with NPR1.

Lines 389-400: the cell death due to AvrRPS4 in Col-0 ecotype is extremely weak as there's no complete receptor complex for this effector. So, one has to use a very high dose to induce cell death in Col-0, certainly higher than the one used for bacterial growth. The authors used the same dose in both assays, so it is likely that what we see as "cell death" is not an effector-triggered response, but rather symptom-associated for the virulent pathogen.

Thank you for pointing this out. Indeed, as the reviewer pointed out, most cell death assays use higher concentrations of Pst DC3000/avrRps4 or Pst DC3000/avrRpt2, but they typically treat Arabidopsis for a relatively short period, usually less than 1 day (Hofius et al., 2009; Zavaliev et al., 2020). In this study, although we used relatively low Pst DC3000/avrRps4 (0.001) injections, we detected cell death under a relatively long period of Pst DC3000/avrRps4 infestation (3 days). Pst DC3000/avrRps4-infested plants multiply significantly in host cells, and therefore we assumed that the propagated pathogens after 3 days of incubation would be sufficient to induce intense cell death. Consequently, we chose this concentration of Pst DC3000/avrRps4 for the experiment.

Lines 407-416: why do you expect "delay of degradation" with autophagy inhibitor? Shouldn't it be the opposite? In Figure S14, if we compare the bands between 120min and 120min+ConA+WM, the effect of autophagy inhibitors is actually quite strong (0.47 vs 0.22), with about 50% more degradation of NPR1 in their presence. So, the conclusion that the degradation of NPR1 is autophagy-independent is wrong according to this result.

Thank you for pointing this out. We have revised the inaccurate description, as outlined in the revised manuscript (lines 413-425).

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